# Recent Progress in Selective Additions of Organometal Reagents to Carbonyl Compounds

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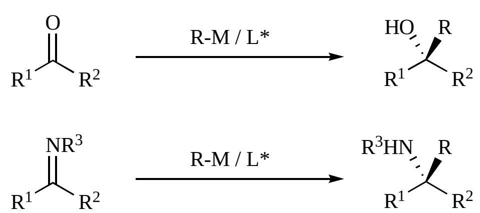
**Abstract:** Carbon–carbon bond forming reactions are simple and direct methods for synthesizing the various skeletons and the diverse structures of organic compounds. Even 100 years after the discovery of the Grignard reaction, organomagnesium, organolithium, and organoaluminum remain popular for carbon–carbon bond formation reactions in modern organic chemistry. Moreover, asymmetric alkyl- or aryl-additions to carbonyl compounds with organometal reagents is attractive due to the enantioselective synthesis of chiral secondary alcohols, which sharply contrasts the enantioselective reduction of carbonyl compounds. Herein, we review recent progress in the selective addition to carbonyl compounds with organometal reagents, particularly Li, Mg, and Zn from both a classic viewpoint using metal ate complexes and from a catalytic enantioselective viewpoint using new chiral ligands.

## 1. INTRODUCTION

For carbon–carbon bond-forming reactions, the addition of carbonyl compounds with organometal reagents is a direct, but versatile method for synthesizing secondary and tertiary alcohols, and even secondary amines [1]. The development of catalytic enantioselective additions to carbonyl compounds is one of the most drastically advancing fields in organic chemistry today (Scheme **1**) [2, 3].




**Scheme 1.** Enantioselective addition of organometal reagents to carbonyl compounds.

It seems that everyday the catalytic enantioselective diethylzinc addition to aldehydes is being improved and that new chiral catalysts are being developed. Oguni and Omi initially established the surprising developments of these catalytic enantioselective additions to carbonyl compounds with organometal reagents, especially diethylzinc to aldehydes [4]. They used a chiral amino alcohol as an asymmetric catalyst for organozinc reagents without additional additives. Later additional metals such as $Ti(OPr^i)_4$ and $Zr(OPr^i)_4$ were shown to increase the catalytic activities and to dramatically improve both the enantioselectivities and yields of the corresponding carbon–carbon bond-forming products [5]. However, modern advanced organic chemistry strongly demands both minimal amounts of reagents/catalysts/solvents/wastes and maximum efficiency in enantioselectivity and chemical yield of products [6]. Certainly, one major advanced strategy in enantioselective addition, particularly in organozinc reagents to aldehydes, is to *never* use other centered metals such as $Ti(OPr^i)_4$, which have been occasionally used in catalytic or large amounts. On the other hand, a lot of good or excellent catalysts for Zn-alkylation to aldehydes are impractical for alkylation of ketones, or even phenylation or alkynylation of aldehydes. Except for the alkylation of aldehydes, the catalytic enantioselective addition of carbonyl compounds is still a challenge and is a topic of a tremendous amount of research. In fact, Ti-activation has been recently reported since ketones and imines have low reactivities relative to aldehydes. In sharp contrast to organozinc reagents [7], other organometal reagents such as organolithium reagents and organomagnesium reagents are less developed, particularly in asymmetric catalysis since their reactivities are too high to be controlled. Therefore, the development of practical asymmetric catalysis and/or a general method to carbonyl compounds are strongly recommended. In this review, recent progress in the selective addition to carbonyl compounds with organometal reagents is discussed with an emphasis on Li, Mg, and Zn from both a traditional viewpoint using metal ate complexes and from a catalytic enantioselective viewpoint using new chiral ligands.

## 2. ATE COMPLEXES DERIVED FROM ORGANOLITHIUM AND ORGANOMAGNESIUM REAGENTS

Recently, ate complexes derived from organolithium, organomagnesium, and organozinc reagents have been in the spotlight as *new* reagents or precursors for the addition to carbonyl compounds [8]. Although catalytic enantioselective addition has yet to be realized, interesting progress in selective halogen-metal exchange reactions or alkyl-selective additions to ketones has been reported. In this section, we review highly effective, new applications of magnesium ate complexes.

### 2.1. Halogen-Magnesium Exchange Reaction with Magnesium Ate Complexes

A half a century after Wittig discovered magnesium ate complexes [9], synthetic applications using magnesium ate

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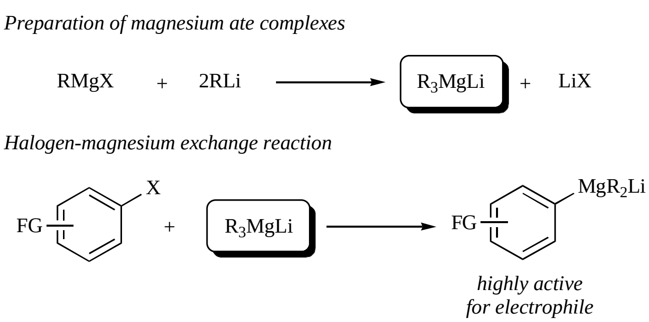
complexes have been limited, but several excellent attempts have reported about the structures of the complexes and synthetic applications in reduction [10]. Recently, Oshima *et al.* made a breakthrough using a magnesium ate complex, which has been demonstrated to be a highly effective reagent for halogen-magnesium exchange reactions [11]. Before then, Knochel reported that even at low temperatures, poly-functional organomagnesium could be efficiently prepared *via* halogen-magnesium exchanges using *i*-PrMgBr [12]. However, magnesium ate complexes are more active reagents than alkylmagnesium halides in halogen-magnesium exchange reactions.

In general, trialkylmagnesium ate complexes ($R_3MgLi$) can be prepared from 1 equiv. of Grignard reagent (RMgX) and 2 equiv. of an alkyllithium reagent (RLi) (Scheme **2**). Oshima *et al*. reported that treating aryl halides with $R_3MgLi$ provides the corresponding organomagnesium reagents in good to excellent yields [13]. These active magnesium ate complexes generated *in situ* and can react with certain electrophiles such as aldehydes, ketones, allyl halides, *etc*.

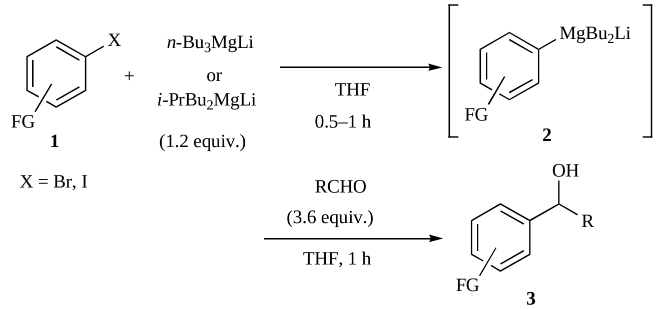



**Scheme 2.** Preparation of magnesium ate complexes and magnesium-halide exchange reaction.

Oshima *et al.* reported that aryl halides (**1**) can be efficiently converted into corresponding magnesium reagents (**2**) by magnesium ate complexes, which subsequently react with aldehydes to give secondary alcohols (**3**) (Table **1**) [11]. *n*-$Bu_3MgLi$ is a good trialkylmagnesate and they found that a mixed magnesate *i*-$PrBu_2MgLi$ is more reactive than *n*-$Bu_3MgLi$ for halogen-metal exchange reactions. For example, *n*-$Bu_3MgLi$ can convert aryl iodides into the corresponding magnesate at –78°C, while the exchange reactions with aryl bromides do not reach completion at –78°C. In contrast, *i*-$PrBu_2MgLi$ can easily exchange with aryl bromides even at –78°C. Advantageously, functional groups such as ethers, esters, amides, and cyano groups remain intact during the mild exchange procedures.

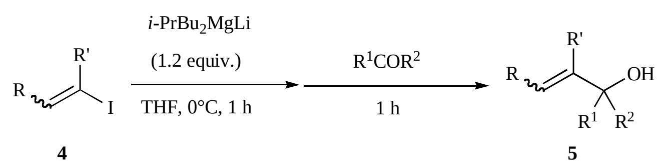
This new procedure using *i*-$PrBu_2MgLi$ is also effective for alkenyl halides (**4**) to prepare alkenylmagnesates, followed by trapping with various electrophiles to give allyl alcohols (**5**) (Table **2**) [11]. Alcohol products *via* a halogen-magnesium exchange reaction and a subsequent reaction with aldehydes or ketones are obtained in good to high yields. Within 1 h the reactivity of *i*-$PrBu_2MgLi$ is high at –78 or 0°C, while the reaction of iodoalkene with *i*-PrMgBr requires a longer reaction time even at room temperature. For *i*-$PrBu_2MgLi$, alkenylmagnesate forms at –78°C even in the presence of a potentially reactive ester moiety. It should be noted that iodine-magnesium exchange reactions with *i*-$PrBu_2MgLi$ proceed without geometrical isomerization in

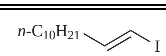
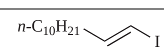
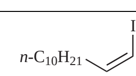
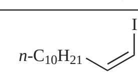
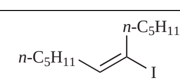
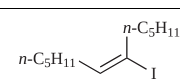
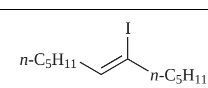
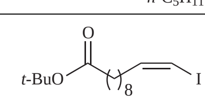
**Table 1. Preparation of arylmagnesates and their reaction with aldehydes**




| Entry | ArX (1) | Reagent | Temp. (°C) | Electrophile | Yield (%) |
|---|---|---|---|---|---|
| 1 | PhI | *n*-$Bu_3MgLi$ | 0 | *n*-$C_6H_{13}CHO$ | 80 |
| 2 | 3-(MeO)$C_6H_4I$ | *n*-$Bu_3MgLi$ | 0 | PhCHO | 93 |
| 3 | PhBr | *n*-$Bu_3MgLi$ | 0 | EtCHO | 87 |
| 4 | 4-($Me_2N$)$C_6H_4Br$ | *n*-$Bu_3MgLi$ | 0 | EtCHO | 94 |
| 5 | 4-$BrC_6H_4Br$ | *i*-$PrBu_2MgLi$ | 0 | EtCHO | 78 |
| 6 | 4-(*t*-$BuO_2C$)-$C_6H_4Br$ | *i*-$PrBu_2MgLi$ | – 78 | *n*-$C_6H_{13}CHO$ | 71 |
| 7 | 4-$NCC_6H_4Br$ | *i*-$PrBu_2MgLi$ | – 78 | *n*-$C_6H_{13}CHO$ | 60 |
| 8 | 2-($Et_2NCO$)-$C_6H_4Br$ | *i*-$PrBu_2MgLi$ | – 78 | EtCHO | 66 |

**Table 2. Preparation of arylmagnesates and their reaction with aldehydes**

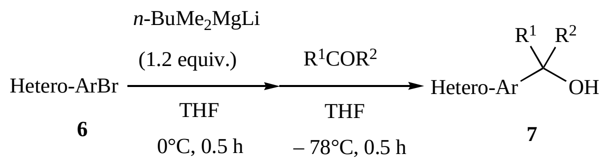



| Entry | Substrate | Temp. (°C) | Electrophile | Yield (%) |
|---|---|---|---|---|
| 1 |  | 0 | PhCHO | 69 |
| 2 |  | 0 | $CH_3COCH_3$ | 75 |
| 3 |  | 0 | PhCHO | 78 |
| 4 |  | 0 | $CH_3COCH_3$ | 66 |
| 5 |  | 0 | EtCHO | 87 |
| 6 |  | 0 | PhCHO | 65 |
| 7 |  | 0 | PhCHO | 70 |
| 8 |  | – 78 | EtCHO | 80 |

the double bond to give the desired carbonyl adducts in good yields.

Oshima *et al.* also established halogen-magnesium exchange reactions for heteroaryl halides (**6**) (Table **3**) [14]. The key to success is a mixed magnesate $n$-BuMe$_2$MgLi since excess butyl groups induce various byproducts when $n$-Bu$_3$MgLi is employed. Bromo- or iodopyridines and bromothiophene are smoothly converted into pyridyl-magnesates or thienylmagnesate *via* a halogen-magnesium exchange reaction and give corresponding alcohols (**7**) in good yields by trapping with excess carbonyl compounds such as aldehydes or ketones.

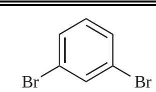
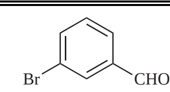
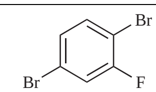
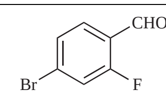
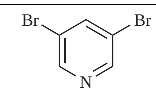
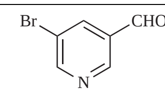
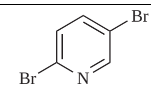
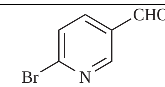
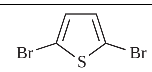
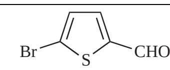
**Table 3. Preparation of heteroarylmagnesates and their reaction with aldehydes and ketones**



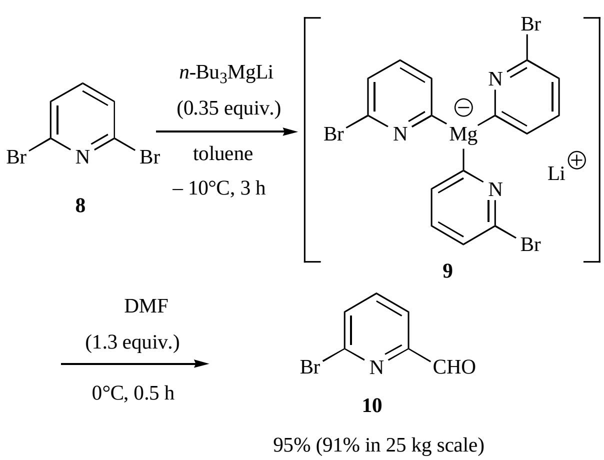

| Entry | ArX (6) | Electrophile | Yield (%) |
|---|---|---|---|
| 1 | 2-PyI | EtCHO | 58 |
| 2 | 2-PyBr | EtCHO | 67 |
| 3 | 3-PyBr | EtCHO | 73 |
| 4 | 3-PyBr | $n$-C$_5$H$_{11}$CO$n$-C$_5$H$_{11}$ | 49 |
| 5 | 3-Thienyl-Br | PhCHO | 78 |

Iida and Mase *et al.* found that $n$-Bu$_3$MgLi is an efficient reagent for the selective mono-bromine-magnesium exchange reaction of 2,6-dibromopyridine **8** [15]. At –10°C the metallated intermediate after halogen-magnesium exchange with 0.35 equiv. of $n$-Bu$_3$MgLi is so stable in toluene that it appears in the structure as **9** (Scheme **3**). A subsequent reaction with DMF proceeds smoothly to give mono-formylated bromopyridine **10** in 95% yield. This indicates that a highly selective mono-formylation *via* magnesium ate complexes is prepared *in situ*. This method has been used to prepare **10** on a 25 kg scale, which shows that it is practical for industrial-scale operations since minimal amounts of $n$-Bu$_3$MgLi are used and cryogenic temperatures are unnecessary. Other dibromoarenes also react with $n$-Bu$_3$MgLi (0.35 equiv.) and subsequent DMF addition selectively affords mono-formylbromoarenes in high yields (Table **4**).

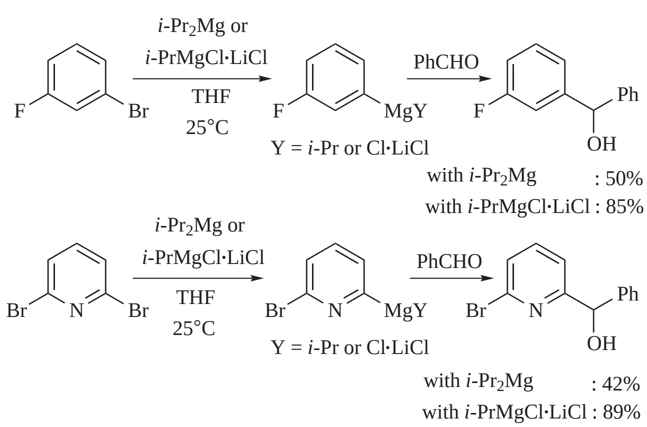
**Table 4. $n$-Bu$_3$MgLi mediated selective mono-formylation of dibromoarenes**

| Entry | Substrate | Product | Yield (%) |
|---|---|---|---|
| 1 |  |  | 99 |
| 2 |  |  | 92 |
| 3 |  |  | 78 |
| 4 |  |  | 71 |
| 5 |  |  | 73 |

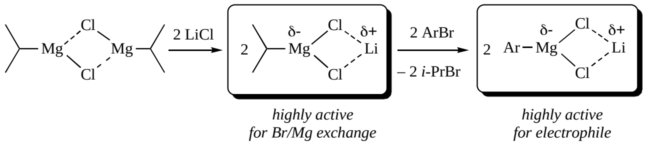
Recently, Krasovskiy and Knochel reported a LiCl-mediated halogen-magnesium exchange reaction and the subsequent reaction with aldehydes as electrophiles [16]. They showed that $i$-PrMgCl·LiCl can be an efficient source to the corresponding arylmagnesium intermediate. The increased activity of $i$-PrMgCl·LiCl sharply contrasts $i$-PrMgCl for arylbromides or heteroarylbromides as shown in Scheme **4**. Interestingly, the presence of LiCl triggers the acceleration of the halogen-metal exchange reaction in the initial stage and the subsequent reaction with the carbonyl compounds in the second stage. They proposed that the increased activity of $i$-PrMgCl·LiCl, is because the addition of LiCl breaks the polymeric aggregates of $i$-PrMgCl, which produces more reactive monomeric complexes (Scheme **5**). Similar salt effects are observed in halogen-zinc exchange reaction where Li(acac) or Cs(acac) show a high performance [17].




**Scheme 3.** Selective mono-bromine-magnesium exchange reaction of 2,6-dibromopyridine.




**Scheme 4.** LiCl-mediated halogen-magnesium exchange and the subsequent reaction with aldehydes.



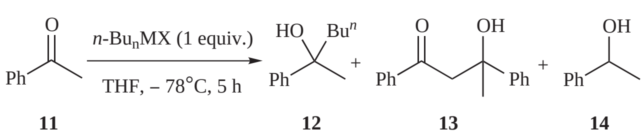
**Scheme 5.** Effect of LiCl in halogen-magnesium exchange.

## 2.2. Alkyl-Selective Addition to Ketones with Magnesium Ate Complexes

While Grignard and organolithium reagents continue to remain popular for C–C bond formation, their reactions with ketones can give rise to reduction products, aldol addition adducts, or starting ketones due to competing enolization problems. Recently, Ishihara and Hatano *et al.* reported the reactions of a range of typical ketones with various magnesium ate complexes [18]. They felt that synthetic applications of magnesium ate complexes are unexplored, particularly in practical reactions such as nucleophiles with carbonyl compounds.

The examination of the *n*-Bu-addition to acetophenone **11** with common lithium and magnesium reagents revealed an inefficient reaction due to a low conversion, a significant amount of aldol (**13**), and a reduced product (**14**). These results imply that to achieve the desired addition to a ketone, the minimum basicity and the maximum nucleophilicity of the alkyl reagent must be carefully controlled. In sharp contrast to these traditional alkyl reagents, namely magnesium ate complex *n*-$Bu_3MgLi$ shows a high reactivity to give Bu-adduct **12** as the sole product in 82% yield (Table **5**).

**Table 5. Addition to acetophenone with magnesium ate complex**

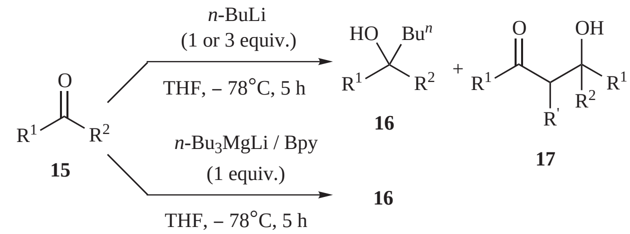


| Entry | *n*-$Bu_nMX$ | Yield (%) | | |
|---|---|---|---|---|
| | | 12 | 13 | 14 |
| 1 | *n*-BuLi | 62 | 7 | 0 |
| 2 | *n*-BuMgCl | 50 | 9 | 8 |
| 3 | *n*-$Bu_3MgLi$ | 82 | 0 | 0 |

Table **6** shows the scope of the reaction with ketones (**15**) and *n*-$Bu_3MgLi$ compared to *n*-BuLi. Even 3 equiv. of *n*-BuLi cannot fully convert the Bu-adduct product (**16**) in several cases. Significantly, a troublesome aldol (**17**) and/or a reduction reaction does not occur under any of the conditions tested in the *n*-$Bu_3MgLi$ systems. Thus, a *n*-$Bu_3MgLi$/Bpy reagent rather than *n*-BuLi or *n*-BuMgCl has an optimal nucleophilicity and a negligible basicity, which makes it ideal for the selective addition to ketones.

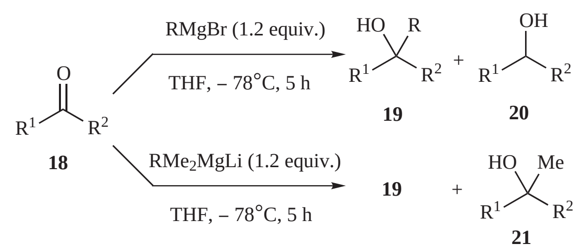
Considering the atom economy of alkylation, the alkyl source (R) in magnesium ate complexes of $R_3MgLi$ is not completely consumed during the reaction. Therefore, *hetero* magnesium ate complexes, *i.e.*, $R^1R^2{}_2MgLi$ ($R^1 \neq R^2$), instead of *homo* magnesium ate complexes such as $R_3MgLi$ are attractive if the second alkyl source ($R^2$) in $R^1R^2{}_2MgLi$ complexes is an unreactive 'dummy'. Ishihara and Hatano *et al.* found that a Me-moiety more favorably yields the $R^1$-selective adduct product as the major product compared to other dummies in $R^1R^2{}_2MgLi$ (*i.e.*, $R^2$ = Me) (Table **7**). In particular, a highly selective ethylation is synthetically important since traditional methods, which commonly use Grignard reaction (EtMgX) or significantly unstable EtLi are limited. In fact, highly alkyl-selective ethylation of ketones (**18**) proceeds quite smoothly with an $EtMe_2MgLi$ ate

**Table 6. Addition to ketones with *n*-$Bu_3MgLi$.**



| Entry | Ketone (15) | Yield (%) of 16 | |
|---|---|---|---|
| | | *n*-BuLi [(equiv.)] | *n*-$Bu_3MgLi$/Bpy |
| 1 | PhCOMe | 77 [3] | 96 |
| 2 | PhCOEt | 90 [1] | 97 |
| 3 | PhCOPr$^i$ | 53 [1] | 80 |
| 4 | α-Naph-COMe | 63 [3] | 71 |
| 5 | β-Naph-COMe | 61 [1] | 93 |
| 6 | PhCOPh | 58 [1] | 80 |
| 7 | α-Tetralone | 52 [3] | 77 |

**Table 7. Highly alkyl-selective addition to ketones with $RMe_2MgLi$ reagents**



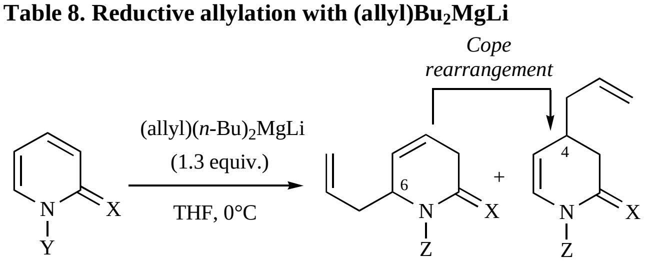
| Entry | Ketone (18) | R | Yield (%) with RMgBr | | Yield (%) with $RMe_2MgLi$ | |
|---|---|---|---|---|---|---|
| | | | 19 | 20 | 19 | 21 |
| 1 | PhCOMe | Et | 71 | 0 | 93 | 6 |
| 2 | PhCOPr$^i$ | Et | 36 | 11 | 92 | 8 |
| 3 | PhCOBu$^t$ | Et | 0 | 50 | 89 | 11 |
| 4 | α-Naph-COMe | Et | 35 | 9 | 94 | 4 |
| 5 | β-Naph-COMe | Et | 66 | 9 | 95 | 5 |
| 6 | α-Tetralone | Et | 36 | 9 | 89 | 7 |
| 7 | 4-PyCOMe | Et | 41 | 48 | 97 | 3 |
| 8 | PhCOPh | Et | 14 | 68 | >99 | <1 |
| 9 | PhCOPh | *i*-Pr | 48 | 36 | 87 | 0 |
| 10 | PhCOPh | *n*-Pr | 15 | 56 | 88 | 12 |
| 11 | PhCOPh | *n*-Bu | 0 | 56 | 92 | 8 |

complex to give the desired Et-adducts **19** in 87–>99% yields. Alkylations with other $RMe_2MgLi$ systems using benzophenone produce highly alkyl-selective additions in high yields. Dramatic results are seen with *n*-$BuMe_2MgLi$ where the yield is improved from 0% with *n*-BuMgBr to 92%.

### 2.3. Selective Allyl-Addition with Magnesium Ate Complexes

Very recently, Sośnicki reported that lithium allyldibutylmagnesate is a powerful allylating reagent for pyridine-2-ones and pyridine-2-thiones (**22**) (Table **8**) [19]. For those compounds, common allylation reagents such as allylmagnesium chloride are ineffective even with a large excess of reagent. In contrast, 6-allyl products (**23**) are selectively obtained from *N*-Me-protected substrates without forming a self-adduct byproduct *via* generating an initial anion at the *N*-Me group [20]. Interestingly, 4-allyl products (**24**) are obtained with perfect regioselectivities *via* a Cope rearrangement after 6-allylations with (allyl)$Bu_2MgLi$.

**Table 8. Reductive allylation with (allyl)$Bu_2$MgLi**




**22** **23** **24**

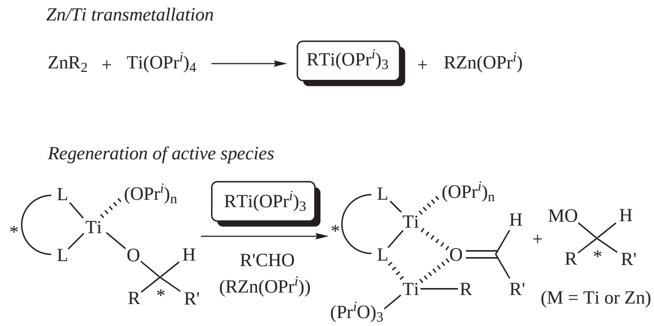
| Entry | X | Y | Z | Yield (%) | Ratio (23:24) |
|---|---|---|---|---|---|
| 1 | O | Me | Me | 61 | 96:4 |
| 2 | S | Me | Me | 80 | 84:16 |
| 3 | O | Li | H | 50 | 0:100 |
| 4 | S | Li | H | 84 | 0:100 |

## 3. ADDITION OF ORGANOZINC REAGENTS

For C–C bond forming reactions with organozinc reagents, $ZnR_2$ is the most popular species, which is easily prepared or commercially available as a solution, neat liquid, and solid [21]. In general, organozinc reagents show little reactivities toward a lot of electrophilic reagents, but this can be controlled by a catalytic amount of a chiral auxiliary without seriously damaging the functional groups. The most competitive catalytic enantioselective reaction in organic synthesis is the addition of diethylzinc to aldehydes since chiral secondary amines are readily obtained in good to excellent yields. Herein, recent progress in asymmetric organozinc addition to aldehydes, ketones, and imines is reviewed.

### 3.1. The Development of Chiral Auxiliaries in Enantioselective Ethylation to Benzaldehyde

To date, $Ti(OPr^i)_4$ has been used in the catalytic enantioselective ethylation to aldehydes. In the presence of equimolar amounts of Ti complexes, there are two key points for a smooth conversion (Scheme **6**): 1) Transmetallation with $ZnR_2$ to generate $RTi(OPr^i)_3$ as an alkylating reagent and 2) regeneration of the active species after C–C bond formation followed by the release of the product and/or the exchange with another $RTi(OPr^i)_3$ [22,23].




**Scheme 6.** The effect of $Ti(OPr^i)_4$.

However, in principle, dialkylzinc addition to aldehydes with *N,O*-ligands should proceed *without* further activation by $Ti(OPr^i)_4$, although a high enantioselectivity is not guaranteed. The reaction is accelerated by an efficient dual activation of the aldehydes (electrophiles) and diethylzinc (nucleophile) with an aminoalkoxy zinc catalyst (see transition states **25** in Fig. (**1**)) [24]. Since the *N,O*-ligand coordinates to the central metal with a coordination style at the *N*-site and has a covalent style at the *O*-site, the chelation of the *N,O*-ligand does not seriously weaken the Lewis acidity of the central metal. This is the reason that *N,O*-ligand catalysis is currently the most established. Of course, even in the presence of $Ti(OPr^i)_4$, the activation of an aldehyde followed by C–C bond formation can proceed in similar transition states (**26** or **27**) or in a dinuclear structure (**28**) [25].

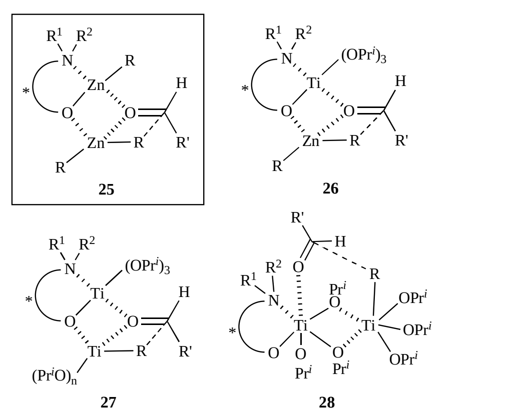


**Fig. (1).** Postulated transition states with *N,O*-ligands.

In sharp contrast to the *N,O*-ligand, the two covalent bonds of the Zn-*O,O*-ligand weaken the Lewis acidity of the central Zn and the aldehyde is not sufficiently activated before C–C bond formation (see **29** in Fig. (**2**)) [26]. Therefore, the strong Lewis acidity of $Ti(OPr^i)_4$ is welcome with the *O,O*-ligands (see **30**–**33**) [27, 28]. Even in the presence of a catalytic amount of $Ti(OPr^i)_4$, Ti may assume a central position as a Lewis acid such as **30** due to its strong oxophilicity.

For other important intermediates, dinuclear or polynuclear Zn or Ti complexes (**34** and **35**) have been proposed

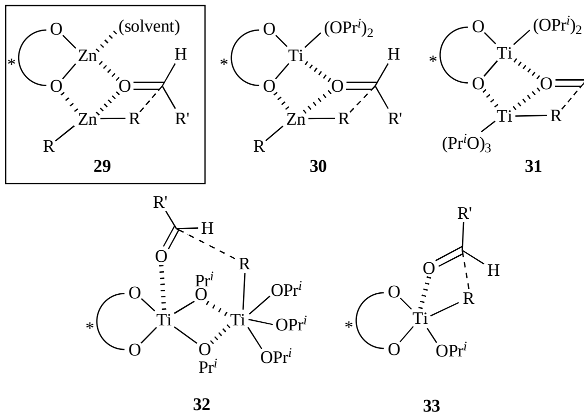
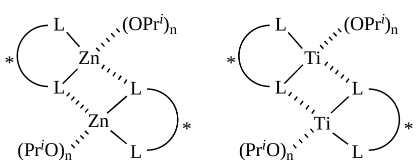



**Fig. (2).** Postulated transition states with *O,O*-ligands.

as the resting state species in Fig. (**3**) [29]. These catalytically inactive species can be observed while initially preparing the chiral ligand and $Ti(OPr^i)_4$ in the absence of aldehydes. In sharp contrast to these dimeric or polymeric species, monomeric resting species can exist by virtue of optional intramolecular coordinations to the central metal. Bis(sulfonamide) ligands by Walsh *et al.* are good examples of monomeric catalysts [30]. The coordination/dissociation of the S=*O* moieties to the centered metal should trigger an active species from the self-inhibiting one (Fig. (**4**)).



**34** **35**

**Fig. (3).** Postulated dinuclear structures.

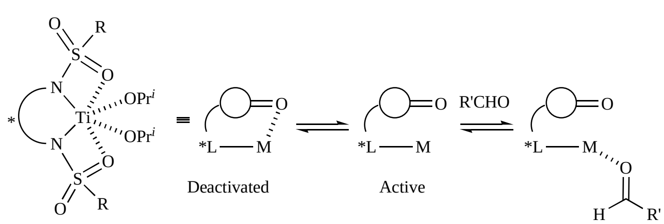



**Fig. (4).** Postulated monomeric structures.

Despite these advantage of Ti complexes [31], modern advanced organic chemistry has preferred to use minimal materials in the concerned reactions. Therefore, the use of $Ti(OPr^i)_4$ is currently not a major method in diethylzinc addition to aldehydes, but is still used for other inert carbonyl compounds such as ketones or imines.

Taking advantage of these frontier situations, we select and review a number of recent (2000–2005), new chiral ligands in Fig. (**5**) except polymer-supported, dendrimer or extremely large ones. They exhibit a *highly enantioselective* ethylation of benzaldehyde in excellent yields *without Ti complexes*, and the results of enantioselectivities are shown in Fig. (**5**) with their catalytic amounts in the parenthesis.

The chiral bidentate (or more) ligands shown here can be divided into four types: *N,O*-ligands, *O,O*-ligands, *N,N*-ligands, and *N,S*-ligands [32]. Among them, the number of *N,O*-ligands is outstanding. Many of the *N,O*-ligands are $C_1$-symmetric, *N,N*-disubstituted amino alcohols and $C_2$-symmetric ligands with double *N,O*-moieties. Moreover, a binaphthyl skeleton with axial chirality is a useful chiral backbone for designing *N,O*-ligands. Ferrocenyl skeletons with planar chirality are unique to *N,O*-ligands. For *O,O*-ligands, the major chiral auxiliaries are $C_2$-symmetric BINOL derivatives, which have di-substitutions in the 3,3-positions in the binaphthyl moieties [33]. Although high enantioselectivities are achieved with as little as 2–5% catalyst loadings, there are very few examples of *N,N*-ligands and *N,S*-ligands from 2000–2005. The catalytic quantity should be addressed to improve asymmetric catalysis. Highly practical ≤1 mol% loadings are limited to 3 examples of *N,O*-ligands, which is *ca.* 5% of all ligands. Surprisingly, even ≤3 mol% loadings are limited to only 14 examples (*ca.* 20% of all). The most common is 5 mol% or 10 mol% loadings and represent as much as *ca.* 60%. Therefore, this area of modern asymmetric catalysis currently consists of a few examples. More research is necessary to further develop this concept.

### 3.2. Enantioselective Diethylzinc Addition to Aliphatic Aldehydes

Despite the hundreds of ligands that have been reported for diethylzinc addition to *aromatic* aldehydes, very few have successfully catalyzed the diethylzinc addition to *aliphatic* aldehydes. In fact, the few effective examples noted that the enantioselective addition to aliphatic aldehydes is difficult, particularly for α-branched aliphatic aldehydes. In this section, we review some recent successful works that afford the Et-adduct products in high enantioselectivities for various aliphatic aldehydes.

Pericàs *et al.* reported the highly enantioselective ethylation of aliphatic aldehydes in the presence of 6 mol% of their devised chiral *N,O*-ligand **36** (Table **9**) [39]. The reactions proceed smoothly in 4 h under the mild conditions and almost quantitatively give the Et-adducts in 90–98% *ee*.

Cho *et al.* reported the enantioselective ethylation to aldehydes with 10 mol% of D-altritol derived *N,O*-ligand **37** (Table **10**) [36]. The corresponding alkyl adducts are obtained with over 80% *ee* and in good to high yields, but relatively high catalytic amounts (10 mol%) of the *N,O*-ligands are required.

Bräse *et al.* demonstrated a remarkably high activity in the enantioselective diethylzinc addition to aliphatic aldehydes with chiral [2.2]paracyclophane-based *N,O*-ligands (**38** and **39**) (Table **11**) [104]. To date, these are some of the most efficient catalysts for the catalytic enantioselective ethylation to aliphatic aldehydes, which include α-branched ones. In fact, the reaction proceeds in excellent yields (100%) with only 1–2 mol% catalyst loadings under mild conditions and without Ti complexes. Extremely high yields and high enantioselectivities are observed for aliphatic aldehydes. For example, a high enantioselectivity (96% *ee*) is obtained for isobutyraldehyde. Interestingly, both **38** and **39** are highly effective, indicating that the central chirality of [2.2]paracyclophane does not affect the absolute stereoselectivity or seriously change the enantioselectivity of the corresponding product.

***N,O*-Ligands**

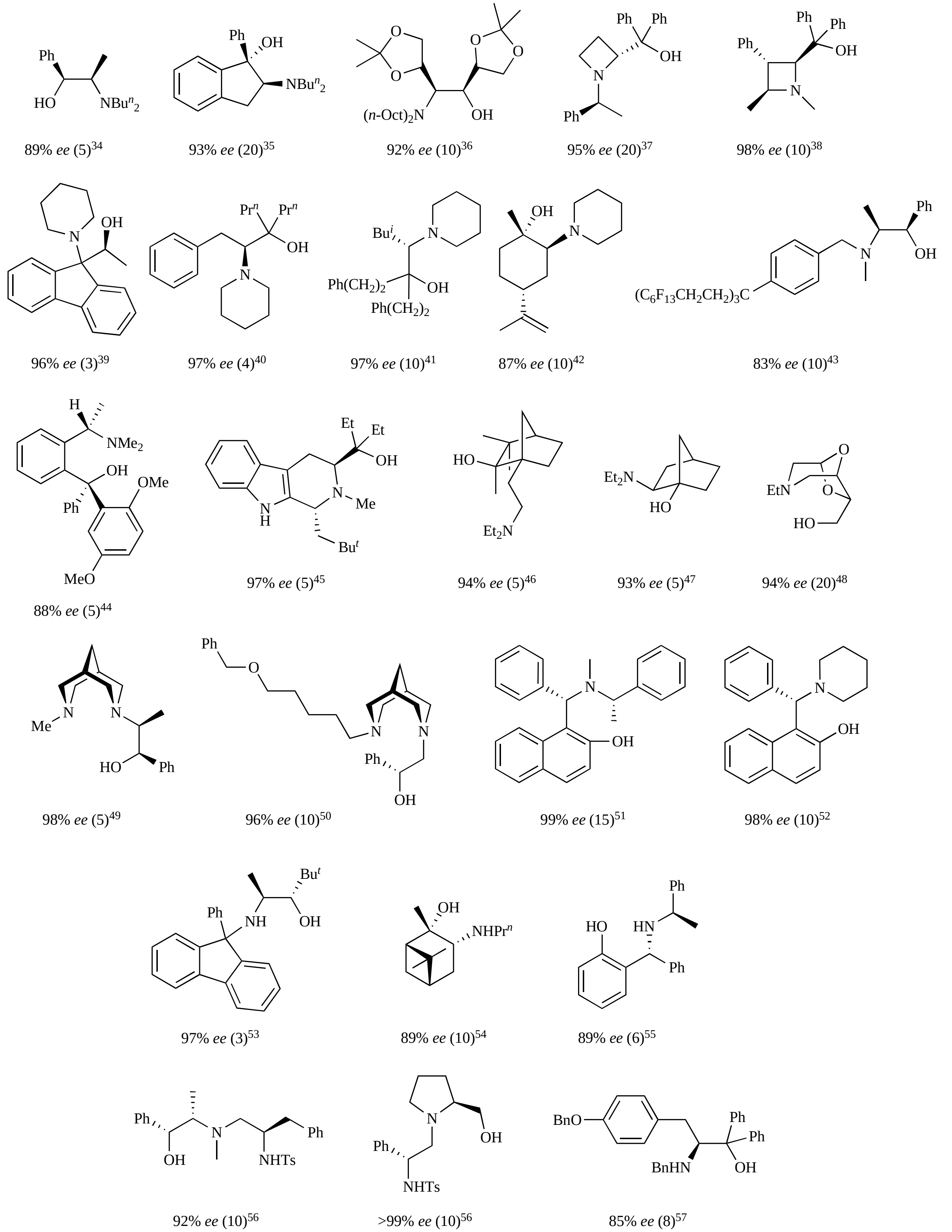
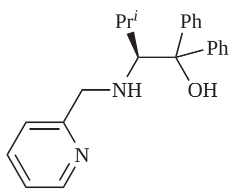


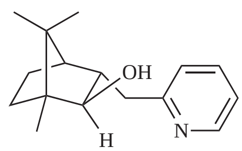

**Fig. (5).** Recent efficient chiral ligands. Data in the parenthesis is the catalytic amounts used for catalysis [34-57].

**Fig. (5) (contd)….**

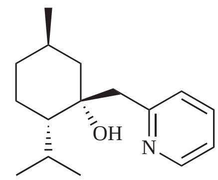
***N,O*-Ligand (continued)**



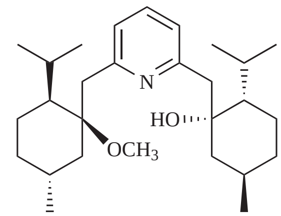
91% *ee* (5)[58]



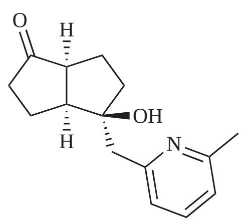
89% *ee* (5)[59]



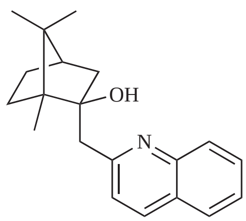
92% *ee* (20)[60]



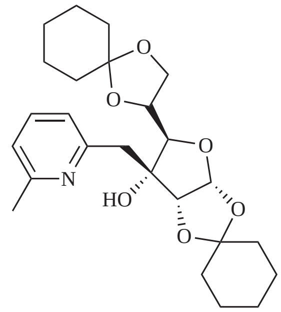
95% *ee* (5)[61]



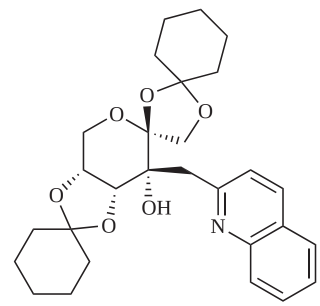
92% *ee* (5)[62]



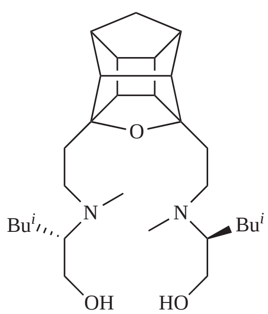
83% *ee* (20)[63]



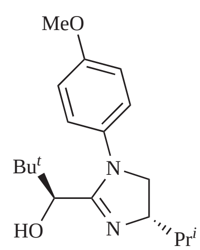
92% *ee* (5)[64]



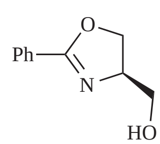
83% *ee* (5)[65]



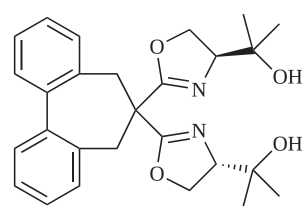
90% *ee* (25)[66]



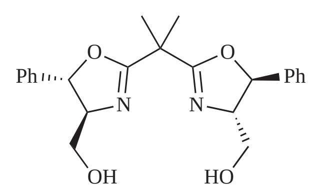
87% *ee* (5)[67]



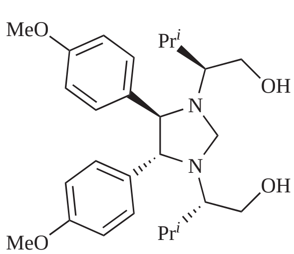
95% *ee* (10)[68]



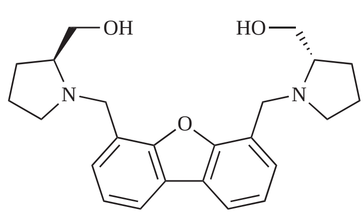
87% *ee* (10)[69]



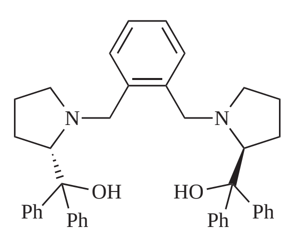
93% *ee* (2)[70]



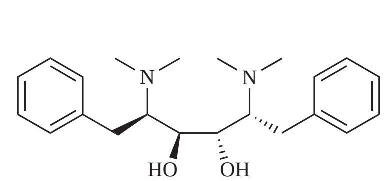
91% *ee* (10)[71]



88% *ee* (2)[72]



95% *ee* (2.5)[73]

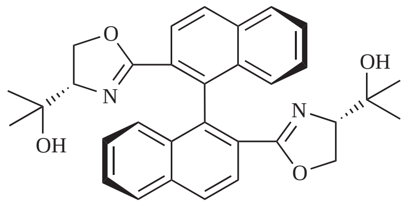


98% *ee* (10)[74]

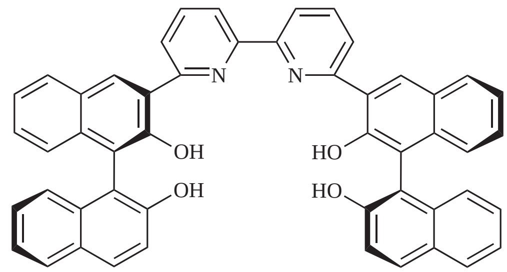
**Fig. (5) (continued).** Recent efficient chiral ligands. Data in the parenthesis is the catalytic amounts used for catalysis [58-74].

**Fig. (5) (contd)….**

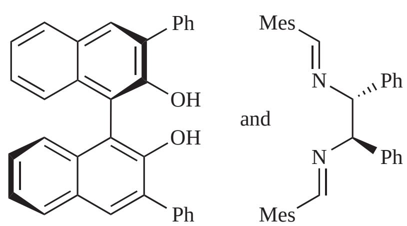
***N,O*-Ligands (continued)**



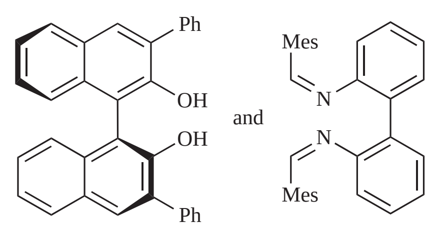
88% *ee* (10)[75]



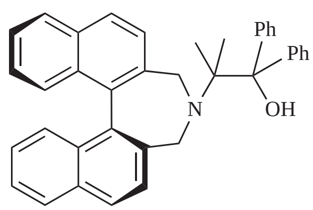
86% *ee* (5)[76]



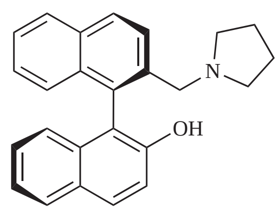
97% *ee* (2)[77]



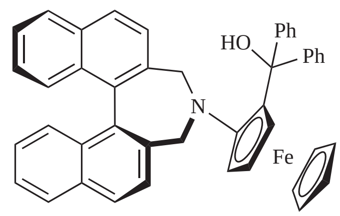
96% *ee* (10)[78]



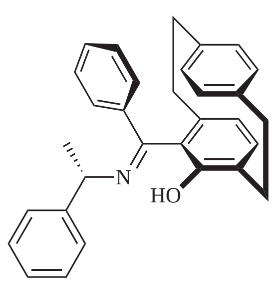
95% *ee* (8)[79]



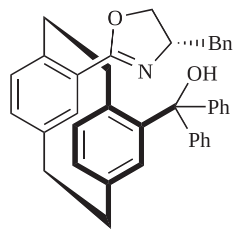
94% *ee* (3)[80]



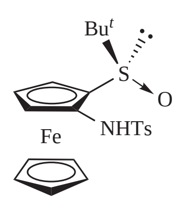
96% *ee* (1)[81]



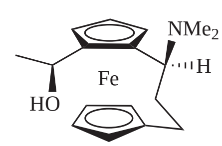
85% *ee* (1)[82]



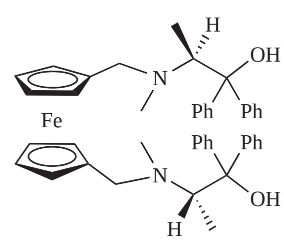
93% *ee* (5)[83]



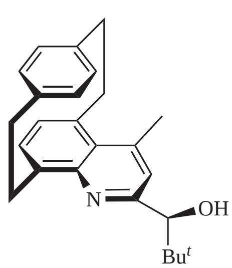
88% *ee* (5)[86]



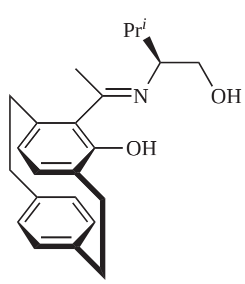
80% *ee* (10)[87]



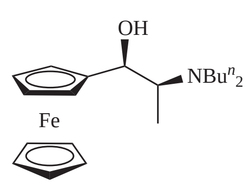
83% *ee* (10)[88]



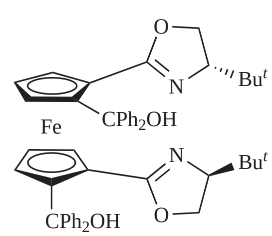
97% *ee* (10)[84]



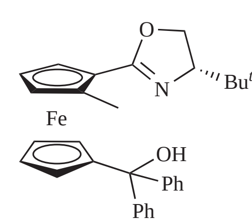
93% *ee* (10)[85]



95% *ee* (10)[89]



93% *ee* (10)[90]

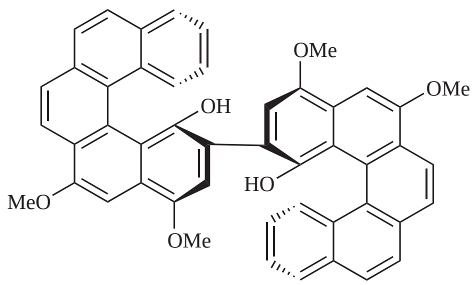


94% *ee* (5)[91]

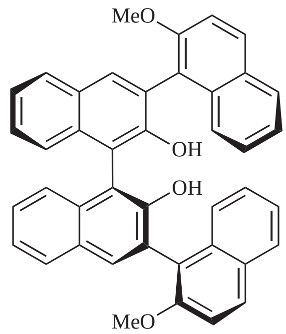
**Fig. (5) (continued).** Recent efficient chiral ligands. Data in the parenthesis is the catalytic amounts used for catalysis [75-91].

**Fig. (5) (contd)….**

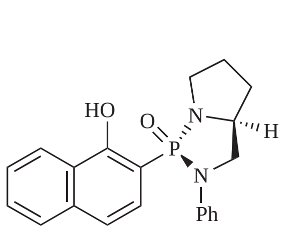
***O,O*-Ligands**



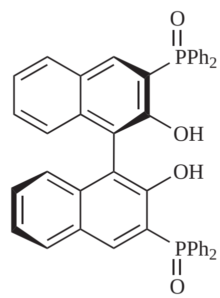
81% *ee* (5)[92]



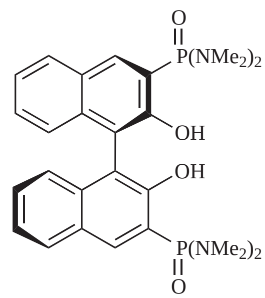
93% *ee* (4.5)[93]



98% *ee* (5)[94]

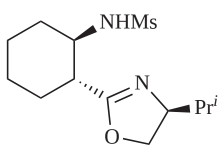


95% *ee* (10)[95]

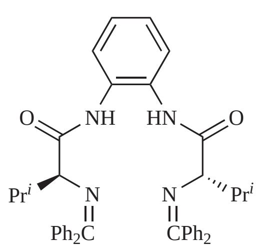


97% *ee* (3)[95]

***N,N*-Ligands**

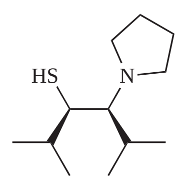


88% *ee* (2)[96]

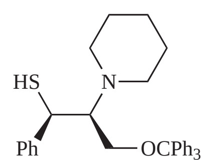


91% *ee* (3)[97]

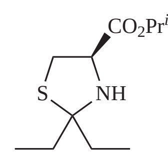
***N,S*-Ligands**



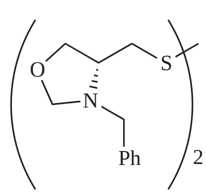
99% *ee* (5)[98]



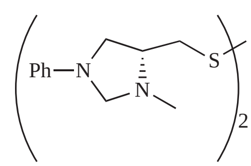
99% *ee* (5)[99]



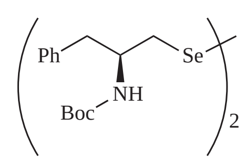
90% *ee* (8)[100]



99% *ee* (2)[101]



91% *ee* (5)[102]



95% *ee* (0.5)[103]

**Fig. (5) (continued).** Recent efficient chiral ligands. Data in the parenthesis is the catalytic amounts used for catalysis [92-103].

Nugent reported that amino alcohol **40** promoted the highly enantioselective addition of diethylzinc to aliphatic aldehydes (Table **12**) [105]. The aliphatic aldehydes are adaptable with 5 mol% of **40** at room temperature and give the corresponding alcohols with 87–99% *ee*. The generality to obtain the products as the almost optically pure compounds (>98% *ee*) is noteworthy. Particularly, this method is applicable to $\alpha$-branched aliphatic aldehydes, which typically are difficult to ethylate.

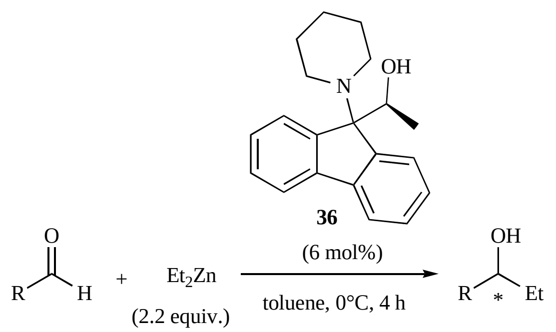
### 3.3. Bifunctional Catalysts in Enantioselective Alkylation to Aldehydes

The tremendous development of *N,O*-ligands reviewed in Fig. (**5**) stems from the nature of the corresponding active Zn complexes with *N,O*-ligands. Using *N,O*-ligands allow the reaction to proceed *via* a dual activation of the aldehyde and diethylzinc without a sophisticated design: the aldehyde is activated by the Lewis acid moiety of the corresponding chiral Zn catalyst, while diethylzinc is activated by the

Lewis base moiety [106]. However, these dual activation points in a chiral Zn complex with a *N,O*-ligand depend on each other since the *N,O*-ligand chelates to the Zn center in a bidentate, tridentate, or tetradentate manner. Therefore, the *N,O*-ligands can be replaced by other types of chiral multi-dentate ligands, in particular, bidentates such as *O,O*-ligands and *N,N*-ligands, and enantioselective alkylation will occur if an independent dual activation is realized. In this section, we review some recent progress from a dual activation viewpoint, namely bifunctional catalysts.

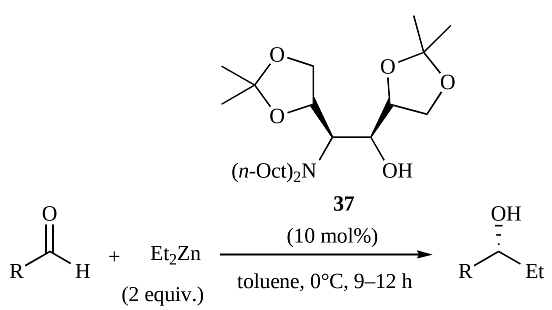
One good example is demonstrated by Dangel and Polt [97]. Their tetracoordinating *N,N*-ligand (**41**) effectively catalyzes the enantioselective alkylation to aldehydes with 3 mol% catalyst loadings (Table **13**). Moreover, dialkylzinc reagents are suppressed to only 1.3 equiv. without Ti reagents. The key is the designed activation of the outer zinc as the alkyl donor toward carbonyl moiety as seen in **42**. Adaptable aldehydes are both aromatic and aliphatic, and the corresponding alkylated products are obtained in high enantioselectivities and high yields.

**Table 9. Enantioselective ethylation of aliphatic aldehydes with *N,O*-ligand 36**



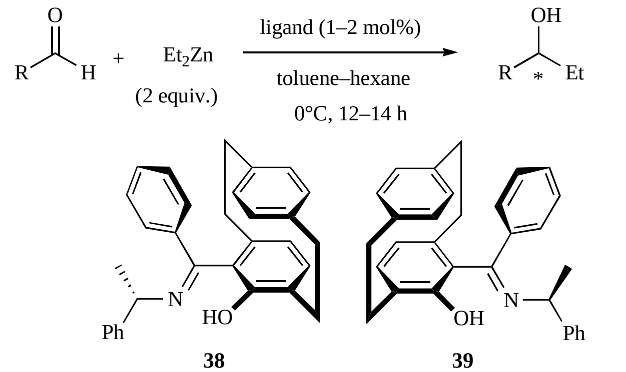

| Entry | R | Yield (%) | *ee* (%) |
|---|---|---|---|
| 1 | $n$-$C_5H_{11}$ | 99 | 91 |
| 2 | $n$-$C_6H_{13}$ | 99 | 91 |
| 3 | $n$-$C_8H_{17}$ | 85 | 90 |
| 4 | *i*-Bu | 95 | 95 |
| 5 | $PhCH_2CH_2$ | 99 | 91 |
| 6 | Cy | 98 | 98 |
| 7 | $Et_2CH$ | 99 | 94 |

**Table 10. Enantioselective ethylation of aliphatic aldehydes with *N,O*-ligand 37.**




| Entry | R | Yield (%) | *ee* (%) |
|---|---|---|---|
| 1 | *n*-Pr | 88 | 82 |
| 2 | $n$-$C_5H_{11}$ | 90 | 84 |
| 3 | $n$-$C_{10}H_{21}$ | 92 | 94 |
| 4 | $PhCH_2CH_2$ | 81 | 90 |
| 5 | *i*-Bu | 92 | 81 |
| 6 | Cy | 78 | 86 |

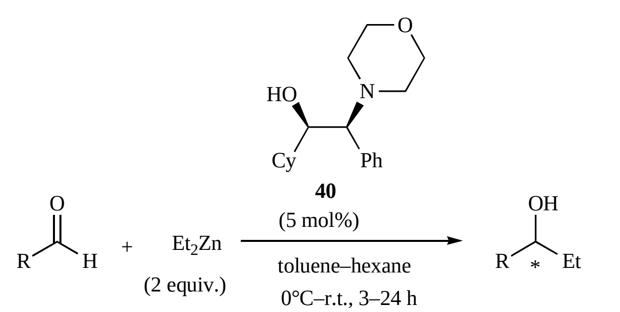
**Table 11. Enantioselective ethylation of aliphatic aldehydes with [2.2]paracyclophane-based *N,O*-ligands**




| Entry | R | Ligand | Yield (%) | *ee* (%) |
|---|---|---|---|---|
| 1 | $Et_2CH$ | **38** | 100 | 97 (+)[a] |
| 2 | $Et_2CH$ | **39** | 100 | 89 (–)[a] |
| 3 | Cy | **38** | 100 | 95 (+)[a] |
| 4 | Cy | **39** | 100 | 95 (–)[a] |
| 5 | $n$-$C_9H_{19}$ | **38** | 100 | 84 (+)[a] |
| 6 | $n$-$C_9H_{19}$ | **39** | 100 | 86 (–)[a] |
| 7 | $n$-$C_5H_{11}$ | **38** | 100 | 86 (+)[a] |
| 8 | $n$-$C_5H_{11}$ | **39** | 100 | 85 (–)[a] |
| 9 | *i*-Pr | **38** | 100 | 96 |
| 10 | *i*-Pr | **39** | 100 | 96 |
| 11 | *t*-Bu | **38** | 97 | 98 (*R*) |

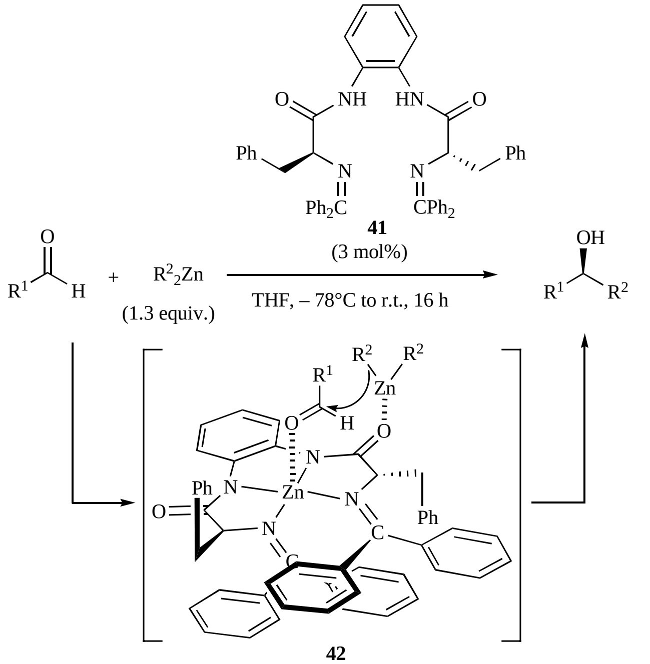
[a]Optical rotation.

**Table 12. Enantioselective ethylation of aliphatic aldehydes with *N,O*-ligand 40**



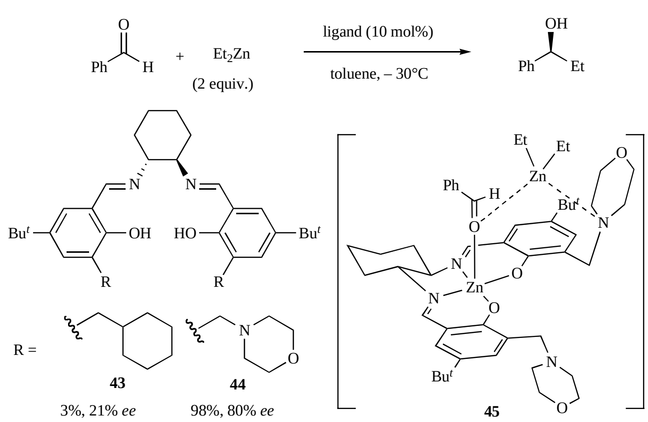

| Entry | R | Yield (%) | *ee* (%) |
|---|---|---|---|
| 1 | *i*-Pr | 93 | 98 |
| 2 | $Et_2CH$ | 91 | 99 |
| 3 | Cy | 93 | 98 |
| 4 | *t*-Bu | 73 | 99 |
| 5 | $CH_2$=$CHCMe_2$ | 78 | 99 |
| 6 | $CH_2$=CMe | 95 | 94 |
| 7 | $n$-$C_5H_{11}$ | 96 | 87 |

**Table 13. Bifunctional catalysis for diethylzinc addition to aldehydes with 41**

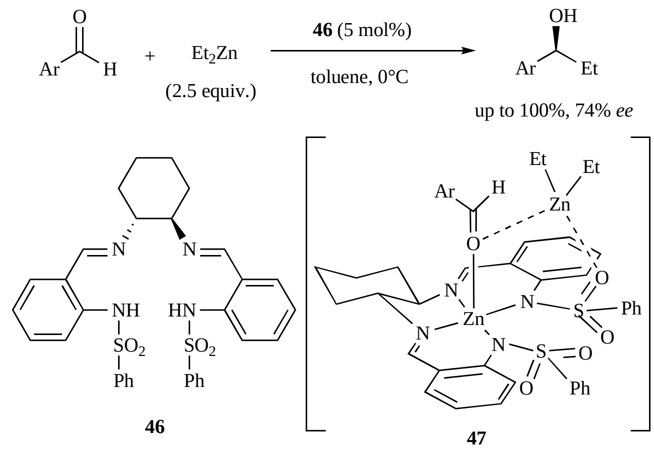



| Entry | $R^1$ | $R^2$ | Yield (%) | *ee* (%) |
|---|---|---|---|---|
| 1 | Ph | Et | 99 | 86 |
| 2 | Ph | Me | 92 | 91 |
| 3 | 2-Furyl | Et | 92 | 86 |
| 4 | $PhCH_2CH_2$ | Et | 81 | 94 |
| 5 | $n$-$C_8H_{17}$ | Et | 99 | 96 |

DiMauro and Kozlowski reported salen-derived catalysts that contain secondary Lewis basic groups for the enantioselective diethylzinc addition without $Ti(OPr^i)_4$. (Scheme **7**) [107]. The centered Zn serves as a Lewis acid and a morpholine moiety serves as a Lewis base, which can independently control the electrophile and nucleophile, respectively (See **45**). It is noteworthy that the catalytic activities and enantioselectivities are greatly enhanced only with bifunctional catalysts (**44**) rather than the mother salen catalysts (**43**) without an independent *N*-moiety of morpholine to serve as a Lewis base moiety.




**Scheme 7.** Bifunctional catalysis for diethylzinc addition to aldehydes.

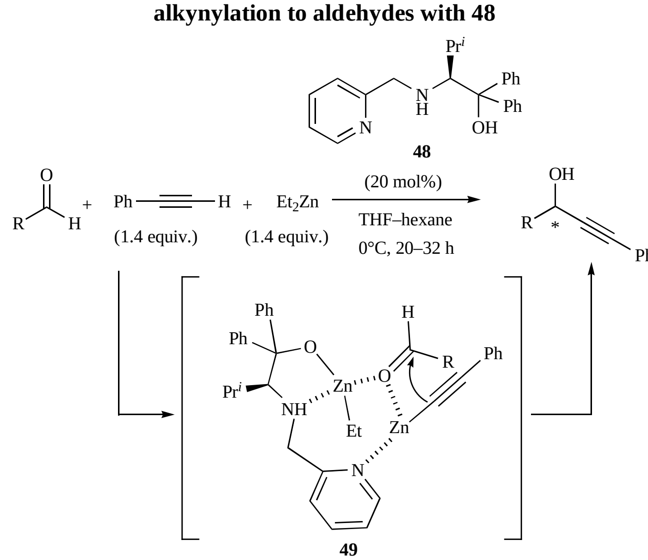



**Scheme 8.** Bifunctional catalysis for diethylzinc addition to aldehydes with **46**.

König *et al*. reported bifunctional disulfonamide catalysts based on cyclohexanediamine for the enantioselective diethylzinc addition to aldehydes without a Ti complex (Scheme **8**) [108]. The Et-adduct product is obtained in 74% *ee* for the enantioselective ethylation of benzaldehyde in the presence of 5 mol% of catalyst **46**. In this case, S=*O* activates the diethylzinc as a Lewis base to ease the alkyl transfer to aldehydes (See **47**).

Wang *et al*. reported a bifunctional catalyst system that enantioselectively alkynylates aldehydes, which shows a more enhanced reactivity than that of the corresponding

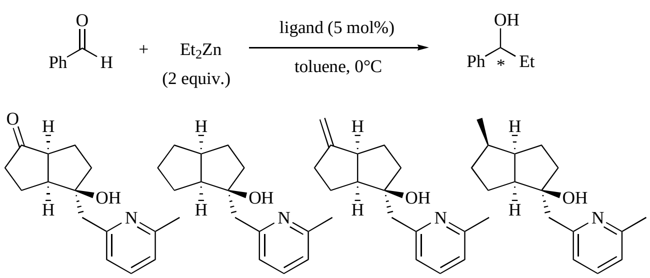
**Table 14. Bifunctional catalysis for enantioselective alkynylation to aldehydes with 48**




| Entry | R | Yield (%) | *ee* (%) |
|---|---|---|---|
| 1 | Ph | 95 | 94 |
| 2 | $3\text{-}MeC_6H_4$ | 85 | 93 |
| 3 | $4\text{-}MeOC_6H_4$ | 86 | 93 |
| 4 | $4\text{-}BrC_6H_4$ | 84 | 87 |
| 5 | $3\text{-}NO_2C_6H_4$ | 80 | 98 |
| 6 | β-Naphthyl | 87 | 91 |
| 7 | *i*-Bu | 91 | 91 |
| 8 | Cy | 88 | 90 |
| 9 | (*E*)-PhCH=CH | 82 | 81 |

amino alcohol ligand without a second activation moiety (Table **14**) [109]. Bifunctional ligand **48** (20 mol%), which is based on an amino alcohol with a pyridyl moiety, can catalyze the addition of phenylacetylene to various types of aldehydes, including aromatic, aliphatic, and $\alpha,\beta$-unsaturated aldehydes *via* **49**, with high enantioselectivities up to 98% *ee*.

Xu and Lin *et al*. designed new bifunctional pyridyl alcohols with a bicyclo[3.3.0]octane scaffold for the ethylation of aldehydes (Scheme **9**) [62]. Other substitutions for the carbonyl group do not effectively increase the enantioselectivities for aldehydes. For example, only **50** affords 3-arylpropanol with >90% *ee*, while other ligands (**51**–**53**) give the same products with <80% *ee*. Therefore, they postulated that carbonyl moiety coordinates with the zinc atom of $Et_2Zn$ and acts as a secondary basic group.




**50** >98%, 92% *ee* **51** >98%, 79% *ee* **52** 88%, 77% *ee* **53** >98%, 77% *ee*

**Scheme 9.** Effect of bifunctional catalyst in enantioselective ethylation of benzaldehyde.

### 3.4. *O,O*-Ligands with Phosphoryl Groups as a New Family of Chiral Auxiliaries

Chiral ligands with phosphoramide or thiophosphoramide have been effective in the asymmetric catalytic alkylation to aldehydes (Fig. (**6**)) [30,110]. However, these chiral auxiliaries require more than a stoichiometric amount of $Ti(OPr^i)_4$ for successful reactions. Recently, in the field of enantioselective alkylation of alde-hydes, new *O,O*-ligands with phosphoryl groups have been developed as a new kind of chiral auxiliary and can provide the alkyl-adducts in high enantioselectivities without Ti complexes.

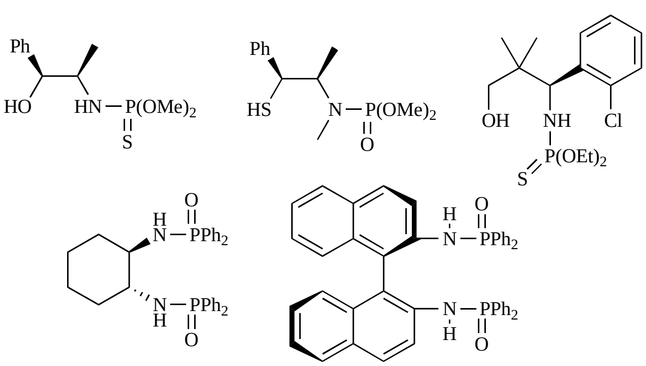
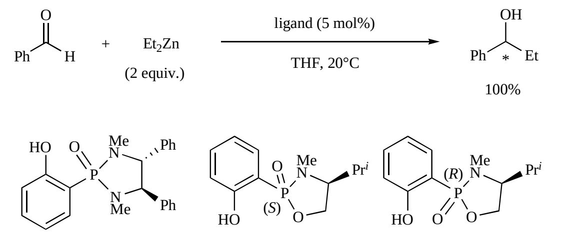
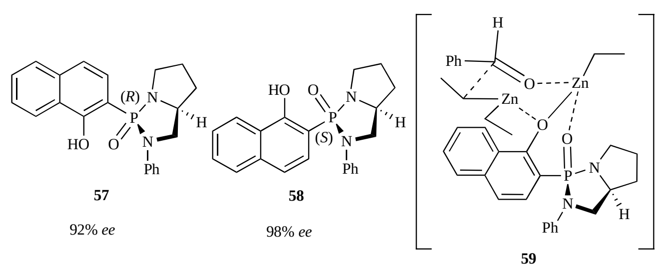


**Fig. (6).** Chiral ligands with phosphoramide groups.

Buono *et al.* developed a highly efficient enantioselective catalysis with a series of chiral (*o*-hydroxyaryl)phosphonamides (**54**–**58**) without Ti complexes (Scheme **10**) [94, 111]. In all cases, benzaldehyde is fully converted to 1-phenylpropanol, but the enantioselectivities vary from 73 to 98% *ee* in the presence of 5 mol% of catalyst and 2 equiv. of $Et_2Zn$. The high activity is remarkable among the *O,O*-ligands reported, although the scope and limitations with other aldehydes were not shown. A plausible mechanism proposes a six-center transition state. EtZn, which is chelated between phenol (OH) and the phosphoryl group (P=O), acts as a Lewis acid to activate the electrophile, while the nucleophile is activated by the phenoxide moiety (See **59**).



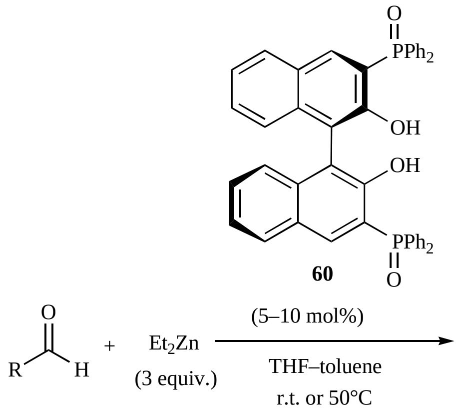

**54** 84% *ee* **55** 73% *ee* **56** 86% *ee*



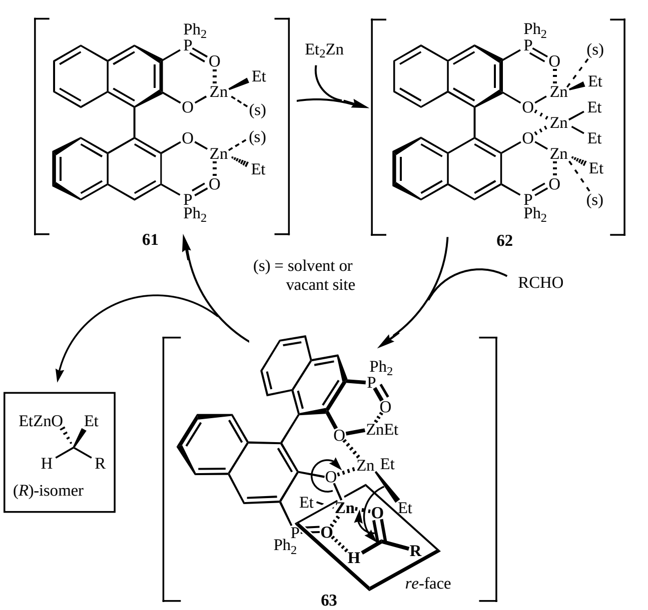

**Scheme 10.** Enantioselective ethylation by *O,O*-ligands with phosphoryl groups.

Ishihara and Hatano *et al.* achieved a highly enantioselective, organozinc addition to aldehydes using new chiral BINOL-Zn catalysts with phosphine oxides [95]. The addition of diethylzinc forms a variety of aldehydes at room temperature in the presence of 10 mol% of **60** as a highly conjugated chiral auxiliary without $Ti(OPr^i)_4$ (Table **15**). Within 12 h Et-adducts are almost quantitatively formed with high enantioselectivities, up to 95% *ee*. Interestingly, the catalytic activity of **60** is dramatically increased at an unprecedented 50°C without a serious loss in enantioselectivity. Moreover, catalyst loading can be decreased to 5 mol% due to the rigid chelation of the P=*O* moiety to the Zn(II) center even under relatively warm conditions. From a mechanistic aspect, **61** initially forms two independent chelations before BINOLate($\kappa^2$*O,O'*) chelation occurs (Scheme **11**). Next the active species (**62**) with a $C_2$-symmetric structure is formed by $Et_2Zn$ coordination to **61**. In their proposed transition states, which include bifunctional moieties, $Et_2Zn$ at the center position is activated by BINOLate($\kappa^2$*O,O'*) as a Lewis base, which acts as the reagent of ethylation, while EtZn at both side positions (*O*=P–C=C–*O*) activates the aldehyde as a Lewis-acid. Postulated transition state **63** is favored since it includes hydrogen bonding between the formic proton and P=O to form a five-membered ring without steric hindrance between the aldehyde and $Et_2Zn$, and electrical repulsion of the lone pairs between P=O and C=O.

**Table 15. Enantioselective ethylation catalyzed by bifunctional BINOL-Zn with phosphineoxides**

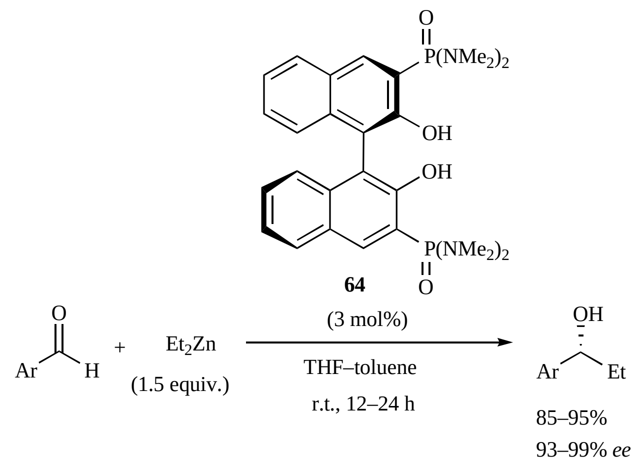


| Entry | R | Temp. | Time (h) | Yield (%) | *ee* (%) |
|---|---|---|---|---|---|
| 1 | Ph | r.t. | 3 | 95 | 95 |
| 2 | 4-$MeC_6H_4$ | r.t. | 8 | >99 | 93 |
| 3 | 4-$PhC_6H_4$ | r.t. | 4 | >99 | 93 |
| 4 | 4-$ClC_6H_4$ | r.t. | 1 | 98 | 94 |
| 5 | 4-$CF_3C_6H_4$ | r.t. | 0.5 | >99 | 94 |
| 6 | 3,4-($OCH_2O$)$C_6H_3$ | r.t. | 4 | 95 | 95 |
| 7 | β-Naphthyl | r.t. | 4 | 95 | 95 |
| 8 | *n*-$C_5H_{11}$ | r.t. | 12 | 71 | 94 |
| 9 | *n*-$C_9H_{19}$ | r.t. | 12 | 72 | 90 |
| 10 | 2-Furyl | r.t. | 3 | 90 | 84 |
| 11 | 3-Thienyl | r.t. | 2 | 94 | 92 |
| 12 | Ph | 50°C | 2 | 84 | 88 |
| 13 | 4-$MeC_6H_4$ | 50°C | 3 | 96 | 88 |
| 14 | 4-$ClC_6H_4$ | 50°C | 0.5 | 92 | 90 |
| 15 | 4-$CF_3C_6H_4$ | 50°C | 0.1 | 98 | 90 |
| 16 | 3,4-($OCH_2O$)$C_6H_3$ | 50°C | 1 | 93 | 89 |
| 17 | β-Naphthyl | 50°C | 1.5 | 92 | 93 |
| 18 | 2-Furyl | 50°C | 1.5 | 93 | 80 |



**Scheme 11.** Proposed catalytic cycle and transition state **63**.

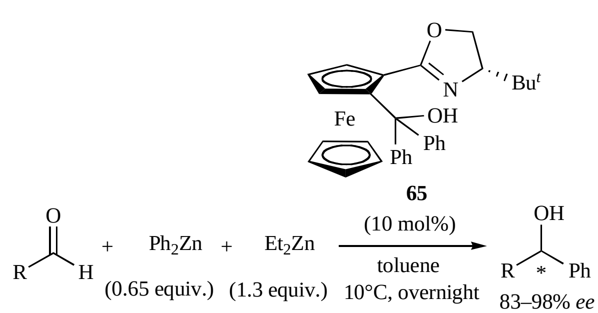
Very recently, Ishihara and Hatano *et al.* devised a new *O,O*-ligand with phosphoramide groups $[P(=O)(NMe_2)_2]$ (**64**) (Scheme **12**) [95]. **64** exhibits a higher catalytic activity than **60** with $[P(=O)Ph_2]$ and the reaction with aromatic aldehydes proceeds using 3 mol% of **64** and 1.5 equiv. of organozinc reagents. The enantioselectivities (93–>99% *ee*) are higher than when using **60** even with lower catalyst loadings and less $Et_2Zn$. Among the *O,O*-ligand category, this example is one of the most remarkable results to date, although applications to aliphatic aldehydes have yet to be realized.



**Scheme 12.** Enantioselective ethylation catalyzed by bifunctional BINOL-Zn with phosphoramides.

### 3.5. Enantioselective Arylation to Aldehydes

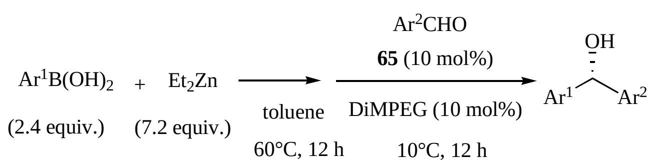
Chiral diaryl methanols can be obtained by the enantioselective reduction of unsymmetrical diaryl ketones or enantioselective C–C bond formation *via* aryl transfer reactions with organometal reagents. However, both methods still have severe limitations in adaptable substrates. In 1999, Haung and Pu reported the first highly enantioselective diphenylzinc additions to aldehydes with chiral BINOL derivatives [112]. On the other hand in 2000, Bolm *et al.* established a highly enantioselective diphenylzinc addition using a 1:2 mixture of $Ph_2Zn$ and $Et_2Zn$ as an organozinc reagent in the presence of ferrocenyl oxazoline alcohol ligand **65** (Scheme **13**) [113]. These advanced and interesting works are introduced in previous reviews [114], but we review recent proceedings around these catalytic enantioselective aryl transfer reactions.



**Scheme 13.** Catalytic enantioselective phenyl transfer to aldehydes with **65**.

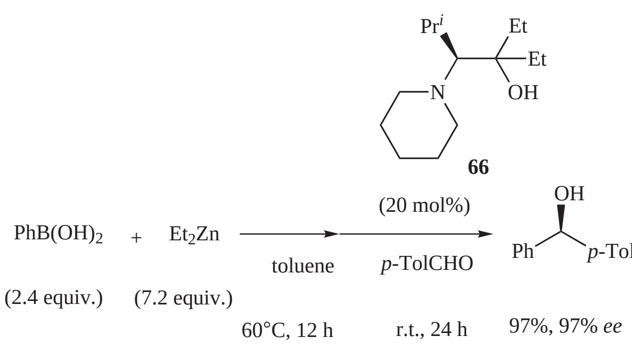
Bolm and Rudolph established a new practical method for synthesizing optically enriched diaryl methanols (Table **16**) [115]. They used readily available arylboronic acids as the aryl sources, which are converted into the active arylzinc species *via* an *in situ* boron-to-zinc exchange reaction in toluene at 60°C in 12 h. This is a great advantage since only phenyl transfer reactions satisfactorily afford arylphenylmethanols with high enantioselectivities. Aryl transfer to aldehydes can proceed with high enantioselectivities and high yields in the presence of 10 mol% of *N*,*O*-ligand **65**. It is noteworthy that both enantiomers of the diaryl methanol products are obtained simply by choosing the appropriate combination of arylboronic acids and aldehydes with a single catalyst loading.

**Table 16. Enantioselective aryl transfer reaction with boronic acids as aryl sources**




| Entry | $Ar^1$ | $Ar^2$ | Yield (%) | *ee* (%) |
|---|---|---|---|---|
| 1 | Ph | $4\text{-ClC}_6H_4$ | 93 | 97 |
| 2 | $4\text{-ClC}_6H_4$ | Ph | 89 | 95 |
| 3 | Ph | $4\text{-PhC}_6H_4$ | 93 | 95 |
| 4 | $4\text{-PhC}_6H_4$ | Ph | 75 | 97 |
| 5 | Ph | $4\text{-MeC}_6H_4$ | 88 | 90 |
| 6 | $4\text{-MeC}_6H_4$ | Ph | 91 | 96 |
| 7 | α-Naphthyl | Ph | 91 | 85 |
| 8 | Ph | Ferrocenyl | 85 | 98 |

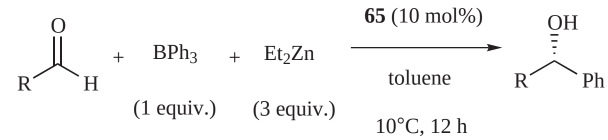
Braga *et al.* adopted another *N*,*O*-ligand **66** for the catalytic enantioselective arylation with arylboronic acids established by Bolm *et al.* [116]. High enantioselectivities are observed in a series of aryl aldehydes.




**Scheme 14.** Enantioselective phenyl transfer reaction with **66**.

Bolm *et al.* reported that the asymmetric phenyl transfer reaction proceeds with an inexpensive triphenylborane as aryl source in the presence of chiral ferrocene based catalyst **65** (Table **17**) [117]. In this method, a long preparation period in the boron-to-zinc exchange reaction is unnecessary, although excess amounts of a Ph-source and Zn-source are required. Remarkable enantioselectivities are achieved in the phenyl transfer to aliphatic and aromatic aldehydes with up to 98% *ee*.

**Table 17. Enantioselective phenyl transfer reaction with triphenylborane**




| Entry | R | Yield (%) | *ee* (%) |
|---|---|---|---|
| 1 | $4\text{-ClC}_6H_4$ | 98 | 97 |
| 2 | $4\text{-MeC}_6H_4$ | 97 | 98 |
| 3 | $4\text{-PhC}_6H_4$ | 88 | 98 |
| 4 | $2\text{-MeOC}_6H_4$ | 89 | 87 |
| 5 | *t*-Bu | 51 | 97 |
| 6 | *n*-$C_6H_{11}$ | 97 | 80 |
| 7 | Cy | 99 | 89 |
| 8 | Et | 97 | 85 |

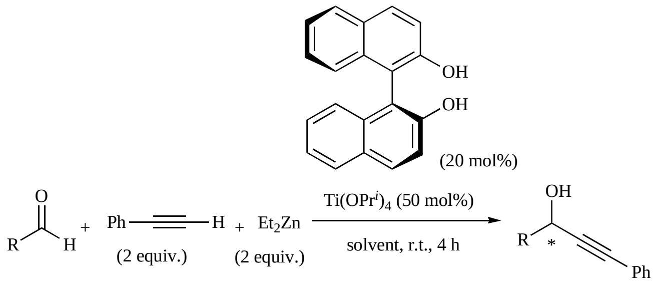
### 3.6. Enantioselective Alkynylation to Aldehydes

Chiral propargylic alcohols are important intermediates in natural products and pharmaceutical compounds. The study of the reactions of alkynylzinc reagents with aldehydes is the most developed among the asymmetric alkyne addition to carbonyl compounds due to both a high reactivity and easy facial enantioselectivity recognition in aldehydes. Unlike lithium acetylides, it is remarkable that zinc acetylides tolerate a wide range of functional groups. In this section, recent topics, which are reported after the precedent comprehensive review [118, 119], are reviewed.

Pu *et al.* demonstrated that readily available and inexpensive (*S*)-BINOL as an *O*,*O*-ligand in combination with $Ti(OPr^i)_4$ is a highly efficient catalyst for the enantioselective alkynylation with various type of aromatic, aliphatic, and α,β-unsaturated aldehydes [120]. Chiral propargyl alcohols are obtained in 4 h in good to excellent yields, 91–99% *ee*, at room temperature in the presence of 20 mol% of the chiral auxiliary (Table **18**). This pioneering work on the catalytic enantioselective alkynylation demonstrated that a variety of aldehydes exhibit a broad scope of high enantioselectivities since previous chiral catalysts do not exhibit such efficiencies. In particular, the availability in aliphatic aldehydes is noteworthy, but 50 mol% of $Ti(OPr^i)_4$ is present.

Pu *et al.* also studied a novel 3,3'-aryl substituted BINOL (**67**) to catalyze the reaction of a terminal alkyne with various aromatic aldehydes to give chiral propargyl alcohols with 80–94% *ee* (Table **19**) [121]. Bulky 3,3'-aryl substituents with adamantyl moieties and a BINOL backbone are key to the high efficiency of the catalyst. Unlike the previously reported BINOL catalysts [122], it is remarkable that this sterically demanding catalyst does not use $Ti(OPr^i)_4$. Moreover, alkylzinc reagents are prepared

**Table 18. Highly efficient BINOL-Ti catalyzed enantioselective alkynylation of a series of aldehydes**



| Entry | R | Solvent | Yield (%) | *ee* (%) |
|---|---|---|---|---|
| 1 | Ph | $CH_2Cl_2$ | 77 | 96 |
| 2 | 3-$ClC_6H_4$ | $CH_2Cl_2$ | 79 | 92 |
| 3 | 4-$ClC_6H_4$ | $CH_2Cl_2$ | 81 | 92 |
| 4 | 2-$MeC_6H_4$ | $CH_2Cl_2$ | 81 | 96 |
| 5 | 3-$MeOC_6H_4$ | $CH_2Cl_2$ | 78 | 93 |
| 6 | 4-$FC_6H_4$ | $CH_2Cl_2$ | 74 | 96 |
| 7 | α-Naphthyl | $CH_2Cl_2$ | 71 | 92 |
| 8 | β-Naphthyl | $CH_2Cl_2$ | 77 | 98 |
| 9 | *n*-$C_8H_{17}$ | $Et_2O$ | 66 | 91 |
| 10 | *n*-$C_7H_{15}$ | $Et_2O$ | 70 | 93 |
| 11 | *n*-Bu | $Et_2O$ | 91 | 93 |
| 12 | *i*-Pr | $Et_2O$ | 84 | 97 |
| 13 | Et | $Et_2O$ | 60 | 94 |
| 14 | Cy | $Et_2O$ | 58 | 95 |
| 15 | $PhCH_2CH_2$ | $Et_2O$ | 99 | 93 |
| 16 | $PhCH_2$ | $Et_2O$ | 93 | 91 |
| 17 | (*E*)-MeCH=CH | $Et_2O$ | 92 | 96 |
| 18 | (*E*)-MeCH=CMe | $Et_2O$ | 93 | 96 |
| 19 | (*E*)-PhCH=CH | $Et_2O$ | 89 | 97 |
| 20 | (*E*)-PhCH=CMe | $Et_2O$ | 96 | 99 |

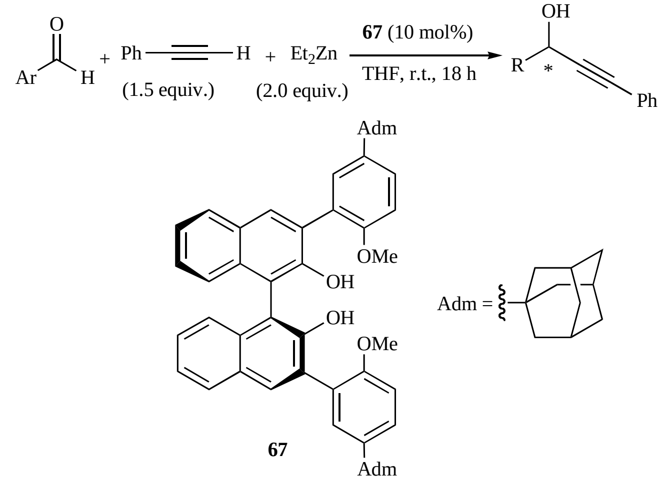
without heating. Also, the experimental procedure for this reaction can be greatly simplified.

Li and Pu reported that BINOL-Salen compound can catalyzed the addition of both aryl- and alkylalkynes to aromatic aldehydes at room temperature with high enantioselectivities (Scheme **15**) [123]. Unlike most other BINOL-based alkynylations, using **68** does not require the addition of $Ti(OPr^i)_4$. But the drawback is some large catalyst loadings based on BINOL.

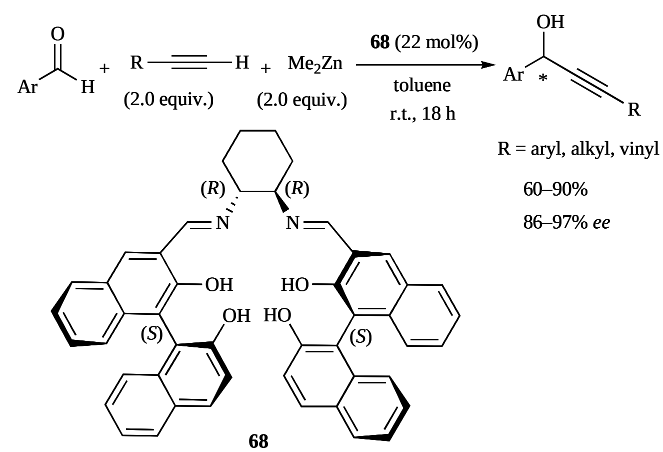
Wang *et al.* reported that a β-sulfonamide alcohol-Ti complex catalyzes the enantioselective addition of phenylacetylene to aromatic aldehydes (Scheme **16**) [124]. 20 mol% of *N,O*-ligand **69** in the presence of 60 mol% of $Ti(OPr^i)_4$ can be employed with numerous aromatic aldehydes and all the corresponding products have high enantioselectivities (>90% *ee*) under mild conditions.

Chan *et al.* achieved interesting improvements in the enantioselectivities of the aklynylation of aldehydes in the presence of phenol and (*R*)-BINOL-Ti catalyst [125]. Interestingly, 10 mol% of additional phenol is an effective activator and better enantioselectivities are observed for a range of aromatic and aliphatic aldehydes (Table **20**).

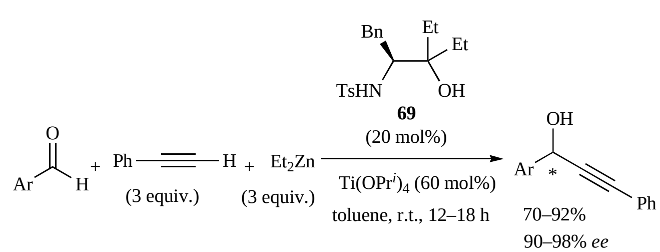
**Table 19. Bulky 3,3'-aryl substituted BINOL-catalyzed enantioselective alkynylation to aldehydes**



| Entry | Ar | Yield (%) | *ee* (%) |
|---|---|---|---|
| 1 | Ph | 75 | 84 |
| 2 | 2-$ClC_6H_4$ | 74 | 94 |
| 3 | 4-$ClC_6H_4$ | 75 | 85 |
| 4 | 2-$MeC_6H_4$ | 72 | 84 |
| 5 | 3-$MeC_6H_4$ | 63 | 84 |
| 6 | 2-$MeOC_6H_4$ | 64 | 91 |
| 7 | 3-$MeOC_6H_4$ | 71 | 85 |
| 8 | α-Naphthyl | 45 | 80 |



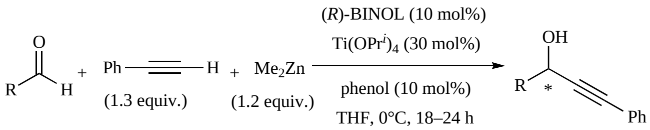
**Scheme 15.** Enantioselective alkynylation catalyzed by BINOL-Salen ligand **68**.



**Scheme 16.** Enantioselective alkynylation catalyzed by *N,O*-ligand **69**.

Particularly, the activation effect is quite significant for *ortho*-substituted benzaldehydes and aliphatic aldehydes.

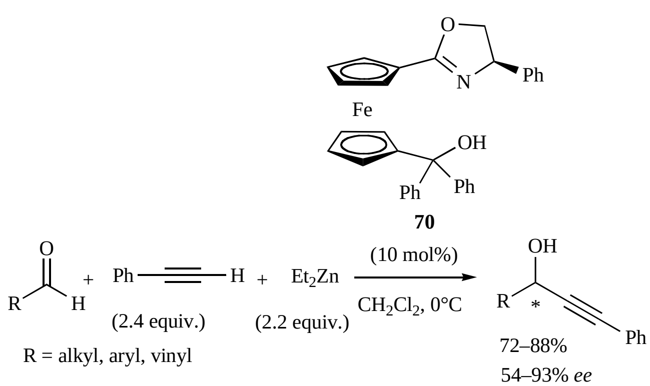
**Table 20. Enantioselective alkynylation catalyzed by BINOL-Ti and phenol**




| Entry | R | Yield (%) | *ee* (%) |
|---|---|---|---|
| 1 | Ph | 85 | 96 |
| 2 | 2-$ClC_6H_4$ | 82 | 88 |
| 3 | 3-$ClC_6H_4$ | 84 | 95 |
| 4 | 4-$ClC_6H_4$ | 83 | 95 |
| 5 | β-Naphthyl | 63 | 94 |
| 6 | Cy | 82 | 86 |
| 7 | *n*-Pr | 91 | 88 |
| 8 | *i*-Pr | 92 | 90 |

The *ee* values improve from 70–80% *ee* with 20 mol% catalyst loading to >85% *ee* with 10 mol% catalyst loading besides phenol, but 30 mol% of a Ti complex are required.

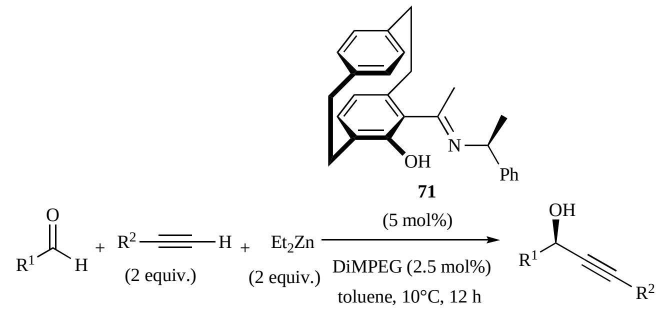
Hou *et al.* reported that ferrocenyl oxazoline alcohol **70** served as a *N,O*-ligand catalyst for the alkynylzinc addition to aldehydes without a Ti complex (Scheme **17**) [126]. Higher enantioselectivities are observed with aromatic aldehydes (67–93% *ee*) rather than aliphatic and α,β-unsaturated aldehydes (54–83% *ee*). The reaction proceeds smoothly under mild conditions to prepare the alkynylzinc reagents without heating.




**Scheme 17.** *N,O*-ferrocene ligand **70** catalyzed enantioselective alkynylation of aldehydes.

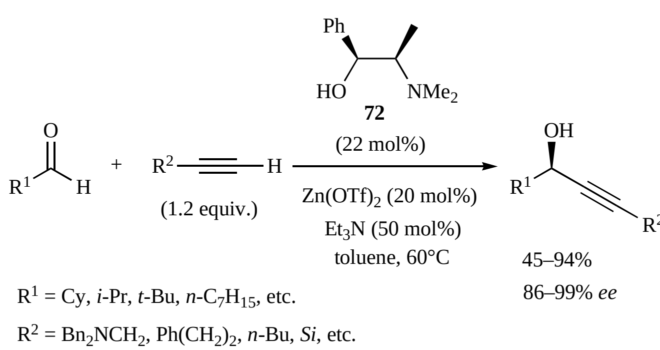
Dahmen established a highly enantioselective alkynylation of aldehydes catalyzed by [2.2]paracyclophane-based catalyst **71** (Table **21**) [127]. It is noteworthy that the reaction with **71** proceeds without Ti complexes in the presence of only 5 mol% catalyst loadings, which is currently superior to conventional amino alcohol ligands. Particularly, high enantioselectivities (up to >98% *ee*) and high yields are observed in the reaction with phenylacetylene and aromatic aldehydes. Other combinations such as benzaldehyde, aliphatic aldehyde, and α,β-unsaturated aldehyde with terminal alkynes are also successful with high enantioselectivities. This method is one of the most practical and synthetically useful catalytic systems thus far.

**Table 21. [2.2]Paracyclophane-based *N,O*-ligand 71 catalyzed highly enantioselective alkynylation**




| Entry | $R^1$ | $R^2$ | Yield (%) | *ee* (%) |
|---|---|---|---|---|
| 1 | Ph | Ph | 90 | 92 |
| 2 | 4-$ClC_6H_4$ | Ph | 89 | 95 |
| 3 | 3-$MeC_6H_4$ | Ph | 92 | 90 |
| 4 | 2-$BrC_6H_4$ | Ph | 86 | >98 |
| 5 | α-Naphthyl | Ph | 86 | 94 |
| 6 | Ph | TMS | 83 | 78 |
| 7 | Cy | Ph | 82 | 77 |
| 8 | Ph | *n*-Bu | 86 | 84 |
| 9 | (*E*)-PhCH=CH | *n*-Bu | 86 | 80 |

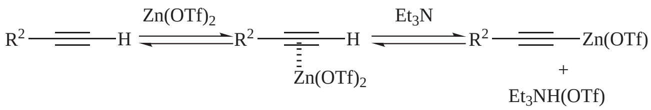
Carreira *et al.* reported that terminal alkynes can undergo an addition to aldehyde with extremely high enantioselectivities using truly catalytic quantities of metal and ligand [128]. For a variety of aliphatic aldehydes and terminal acetylenes, the reaction proceeds within 24 h in the presence of $Zn(OTf)_2$ (20 mol%), $Et_3N$ (50 mol%), and chiral *N,O*-ligand **72** (22 mol%) in toluene at 60°C (Scheme **18**). The corresponding chiral propargyl alcohols are obtained in >80% yield and >90% *ee* in almost all cases. In this method, a catalytic amount of Zn-acetylide, which can be generated in the equilibrium *via* a π-complex of $Zn(OTf)_2$ followed by deprotonation of the terminal acetylene in the presence of a mild base such as $Et_3N$ (Scheme **19**). Thus, the process epitomizes atom economy that is expected in modern chemistry.




**Scheme 18.** $Zn(OTf)_2$ catalyzed enantioselective addition of acetylene to aldehydes with **72**.

## 3.7. Enantioselective Alkylation and Arylation to Ketones

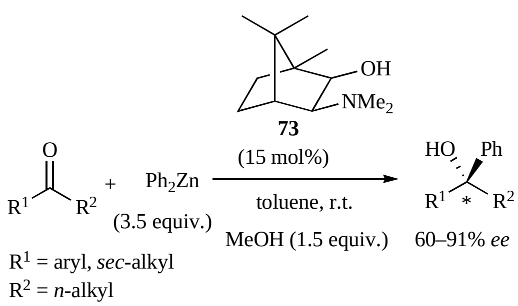
The catalytic enantioselective addition of organometal reagents to ketones is still a challenging field [129,130].



**Scheme 19.** Generation of zinc alkynylides.

Although effective methods have been developed for the catalytic enantioselective reduction of ketones, there are few catalytic enantioselective C–C bond forming reactions due to their low reactivities and the difficulties in controlling the facial stereoselectivity. In particular, organozinc reagents are less reactive than organolithium or organomagnesium reagents, but organozinc reagents are preferred since they tolerate the presence of many functional groups. Despite these difficulties, the enantioselective addition of organozinc reagents to ketones is rapidly advancing, especially in catalytic enantioselective alkynylations and we review the recent fruitful progress.

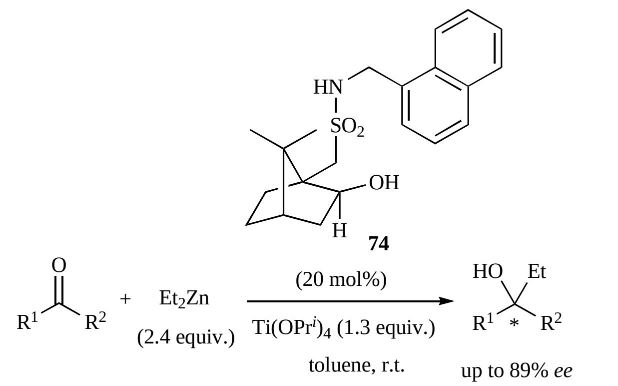
Dosa and Fu published the first catalytic enantioselective addition of organozinc reagents in 1998 [131]. In this pioneering work, diphenylzinc addition to ketones is promoted by 15 mol% of DAIB (**73**) and the corresponding chiral benzyl alcohols are obtained in good to high enantiomeric excesses up to 91% *ee* (Scheme **20**).



**Scheme 20.** The first catalytic enantioselective aryl transfer reaction with **73**.

Around the same time, Ramón and Yus reported the first catalytic enantioselective addition of alkylzinc reagents to ketones [132]. Although an equimolar amount of a Ti complex is required at room temperature, some substrates exhibit high enantioselectivities, up to 89% *ee*, in the presence of chiral sulfonamide ligand (**74**), which is based on camphor with a hydroxyl group (Scheme **21**).

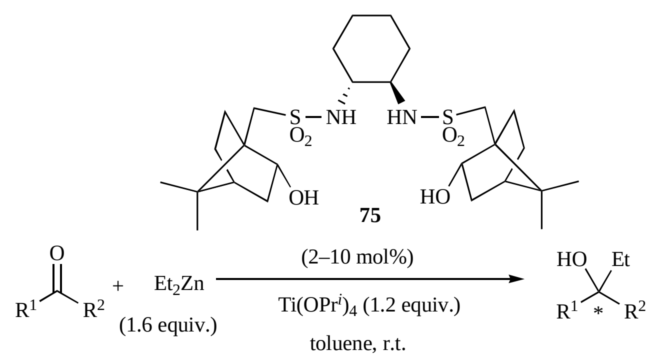
Walsh *et al.* employed $C_2$-symmetric bis(sulfonamide) diol **75** based on camphor and 1,2-cyclohexanediamine [133]. This ligand is an excellent chiral auxiliary for highly enantioselective dialkylzinc addition to a series of aromatic and aliphatic ketones (Table **22**). This is one of the most effcient catalysts and even more attractive is that only 2 mol% catalyst loading is necessary, but on the downside, an equimolar amount of $Ti(OPr^i)_4$ is required. Moreover, this catalyst can be applied to a large-scale synthesis. For example, 2 mol% of **75** catalyzes the enantioselective Et-addition of 3-methylacetophenone (5.0 g) to give the corresponding tertiary alcohol with perfect enantioselectivity (99% *ee*).



**Scheme 21.** The first catalytic enantioselective alkyl transfer reaction with **74**.

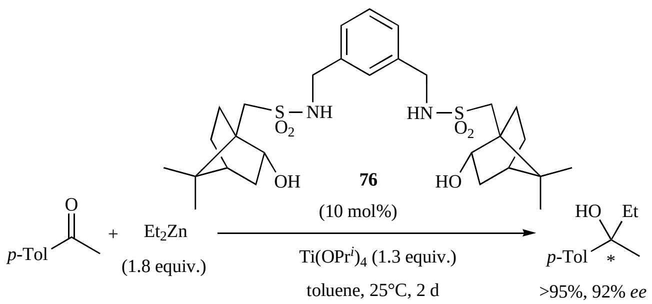
Yus *et al.* and Walsh *et al.* independently devised chiral ligand **75** [134]. Yus and coworkers determined that a $C_2$-symmetric bis(sulfonamide) diol (**76**) is effective, but 10 mol% catalyst loading and an equimolar amount of a Ti complex are required (Scheme **22**). The generality for ketones is lower than **75** [135].

**Table 22. Catalytic enantioselective addition of $Et_2Zn$ to ketones with 75**



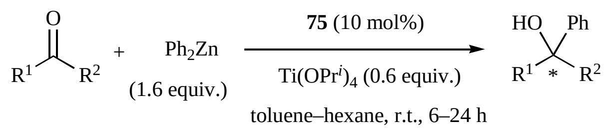
| Entry | Ketone | 75 (mol %) | Yield (%) | *ee* (%) |
|---|---|---|---|---|
| 1 | PhCOMe | 2 | 71 | 96 |
| 2 | 3-$MeC_6H_4$COMe | 2 | 78 | >99 |
| 3 | $\alpha$-Tetralone | 10 | 35 | >99 |
| 4 | $PhCOBu^t$ | 2 | 79 | 88 |
| 5 | (*E*)-PhCH=CHCOMe | 2 | 80 | 90 |
| 6 | $PhCH_2CH_2$COMe | 10 | 68 | 70 |
| 7 | 1-CyclohexenylCOMe | 2 | 56 | 96 |

García and Walsh also reported that a $C_2$-symmetric bis(sulfonamide) diol (**75**) catalyzes a highly enantio-selective phenylation (Table **23**) [136]. Significant improvements like a low catalyst loading of 10 mol%, and only 1.6 equiv. of $Ph_2Zn$ are observed from the first example by Dosa and Fu. Extremely high enantioselectivities and chemical yields are shown in a variety of aromatic and aliphatic ketones, and $\alpha,\beta$-unsaturated enones. The wide scope of this method using a readily available ligand is attractive from a practical asymmetric catalysis viewpoint, although 60 mol% of $Ti(OPr^i)_4$ is used.



**Scheme 22.** Catalytic enantioselective addition of $Et_2Zn$ to ketones with **76**.

**Table 23. Catalytic enantioselective addition of $Ph_2Zn$ to ketones with 75**



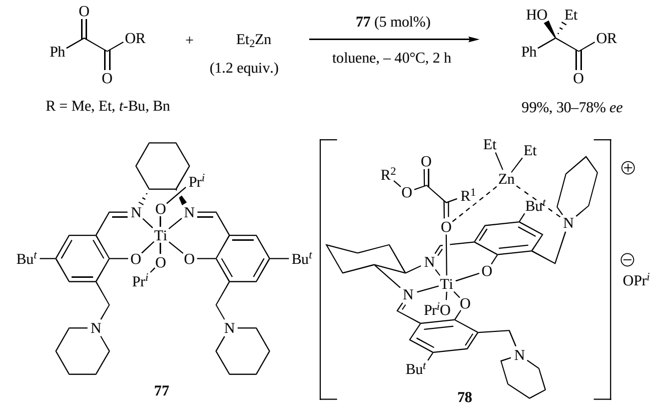
| Entry | Ketone | Yield (%) | *ee* (%) |
|---|---|---|---|
| 1 | 3-$ClC_6H_4COEt$ | 99 | 92 |
| 2 | 3-$CF_3C_6H_4COMe$ | 93 | 95 |
| 3 | 2-$BrC_6H_4COMe$ | 76 | 95 |
| 4 | β-NaphCOMe | 99 | 96 |
| 5 | (*E*)-PhCH=CHCOMe | 58 | 91 |
| 6 | CyCOMe | 81 | 87 |
| 7 | 1-CyclohexenylCOMe | 94 | 93 |

DiMauro and Kozlowski reported the first catalytic enantioselective addition of organozinc reagents to α-ketoesters (Scheme **23**) [137]. α-Ketoester is regarded as an activated ketone, thus, a high reactivity is expected. They designed a bifunctional amino salen-Ti catalyst (**77**) with a Lewis-acid moiety in the Ti center to activate ketoester and Lewis-base centers in the piperidine tethers to activate the organozinc reagent (See **78**). The reactions proceed in toluene at –40°C in 2 h without further additions of $Ti(OPr^i)_4$, except the 5 mol% of the centered Ti. The resultant α-alkyl-α-hydroxyesters are obtained in high yields, but with low to moderate enantioselectivities (up to 78% *ee*).

### 3.8. Enantioselective Alkynylation to Ketones

Cozzi achieved the first general catalytic enantioselective addition of terminal alkynes to unactivated ketones (Table **24**) [138]. To overcome the low reactivity of the substrates, the concept of double activation (*i.e.,* Lewis acid-base) is involved in the catalytic methodology, especially the use of salen complexes (See **80**). For a range of aromatic and aliphatic ketones, 20 mol% of (*R*,*R*)-*N*,*N*'-bis(3,5-di-*tert*-butylsalicilidene)-1,2-cyclohexanediamine (**79**) in toluene at room temperature effectively provides the adduct products with moderate enantioselectivities.

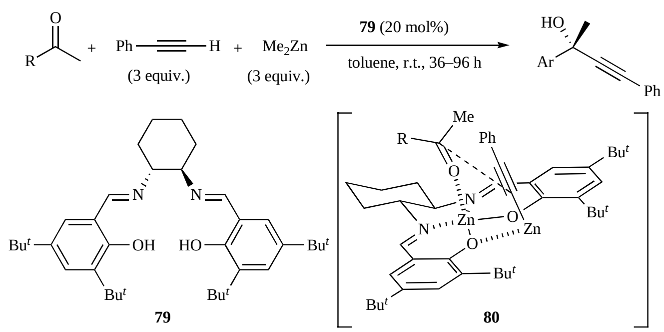
Katsuki and Saito reported the Zn(salen)-catalyzed asymmetric alkynylation of ketones [139]. They used 8 mol% of an *in situ* generated Zn complex with chiral salen ligand **81** with axially chiral binaphthyl units (Table **25**). For



**Scheme 23.** The first catalytic enantioselective alkyl transfer reaction to α-ketoesters with **77**.

a wide range of ketones and terminal acetylenes, the alkynylated products are obtained in moderate to good yields with high enantioselectivities.

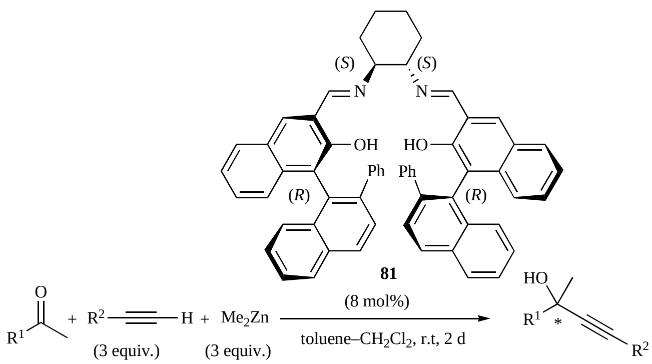
**Table 24. Catalytic enantioselective alkynylation to ketones with 79**



| Entry | R | Yield (%) | *ee* (%) |
|---|---|---|---|
| 1 | Ph | 72 | 61 |
| 2 | 2-$BrC_6H_4$ | 50 | 70 |
| 3 | *t*-$BuC_6H_4$ | 45 | 66 |
| 4 | 3-$CF_3C_6H_4$ | 81 | 61 |
| 5 | *i*-Pr | 84 | 57 |
| 6 | *t*-Bu | 89 | 80 |

Chan *et al.* established a highly enantioselective addition of phenylacetylene to aromatic ketone with camphor-sulfonamide ligand **74** [140], which was devised by Yus *et al.* [132]. A variety of aromatic ketones with phenyl-acetylene in the presence of $Me_2Zn$ are catalyzed by 10 mol% each of $Cu(OTf)_2$ and the camphorsulfonamide ligand in dichloromethane at 0°C for 48 h. The corresponding chiral propargyl alcohols are obtained in good to high yields and excellent enantiomeric excesses up to 97% *ee* (Table **26**). It is noteworthy that Chan's examples are highly efficient with regard to a low catalyst loading and excellent enantio-selectivities since selective addition to ketones is difficult. Substitution at the *ortho* position of the substrate favorably affected the enantioselectivities since the steric hindrance of

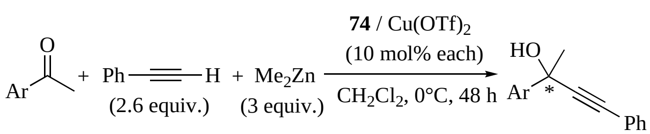
**Table 25. Catalytic enantioselective alkynylation to ketones with 81**




| Entry | R¹ | R² | Yield (%) | *ee* (%) |
|---|---|---|---|---|
| 1 | 4-$ClC_6H_4$ | Ph | 70 | 84 |
| 2 | 4-$NO_2C_6H_4$ | Ph | 86 | 77 |
| 3 | *i*-Pr | Ph | 60 | 71 |
| 4 | *t*-Bu | Ph | 75 | 91 |
| 5 | *t*-Bu | 4-$CF_3C_6H_4$ | 67 | 92 |
| 6 | *t*-Bu | 4-$MeOC_6H_4$ | 76 | 93 |

the *ortho* substituents restricts the orientation of the substrates.

**Table 26. Catalytic enantioselective alkynylation to ketones with 74**

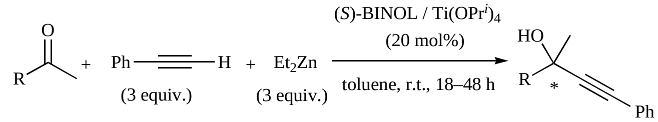



| Entry | Ar | Yield (%) | *ee* (%) |
|---|---|---|---|
| 1 | Ph | 92 | 88 |
| 2 | 2-$BrC_6H_4$ | 65 | 96 |
| 3 | 3-$BrC_6H_4$ | 80 | 82 |
| 4 | 4-$BrC_6H_4$ | 75 | 91 |
| 5 | 2-$ClC_6H_4$ | 94 | 97 |
| 6 | 2-$FC_6H_4$ | 91 | 96 |
| 7 | 2-$MeC_6H_4$ | 49 | 96 |

Wang *et al.* demonstrated that a (*S*)-BINOL ligand combined with a catalytic amount of $Ti(OPr^i)_4$ effectively catalyzes the enantioselective addition of alkynylzinc to unactivated ketones [141]. A 1:1 ratio of (*S*)-BINOL and $Ti(OPr^i)_4$ is critical to activate inert ketones so that the corresponding highly acidic (BINOLate)-$Ti(OPr^i)_2$ is formed. Good to excellent enantioselectivities and chemical yields are observed for aromatic and aliphatic ketones (Table **27**). Around the same time, Cozzi reported the BINOL-Ti catalyzed asymmetric addition, but used 1.5 equiv. of $ClTi(OPr^i)_3$ to prepare titanium phenylacetylene [142].

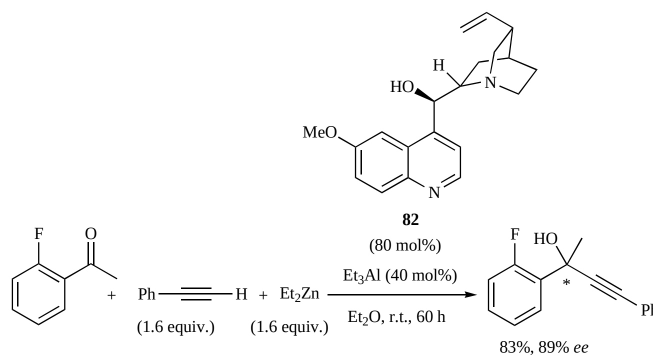
Very recently, Wang *et al.* reported the enantioselective phenylacetylene addition to aromatic ketones catalyzed by cinchona alkaloid-aluminum complexes **82** (Scheme Scheme) [143]. This is the first chiral Al-catalysis that allows the enantioselective alkynylzinc addition to un-

**Table 27. (*S*)-BINOL-Ti catalyzed asymmetric addition of phenylacetylene to ketones**




| Entry | R | Yield (%) | *ee* (%) |
|---|---|---|---|
| 1 | Ph | 67 | 85 |
| 2 | 3-$MeOC_6H_4$ | 81 | 92 |
| 3 | 4-$MeC_6H_4$ | 64 | 87 |
| 4 | 4-$ClC_6H_4$ | 73 | 89 |
| 5 | *i*-Bu | 91 | 63 |
| 6 | (*E*)-PhCH=CH | 88 | 73 |

activated ketones to proceed with 70–89% *ee* under optimized conditions. However, both the amount of Al catalyst and cinchona alkaloid are inappropriate.

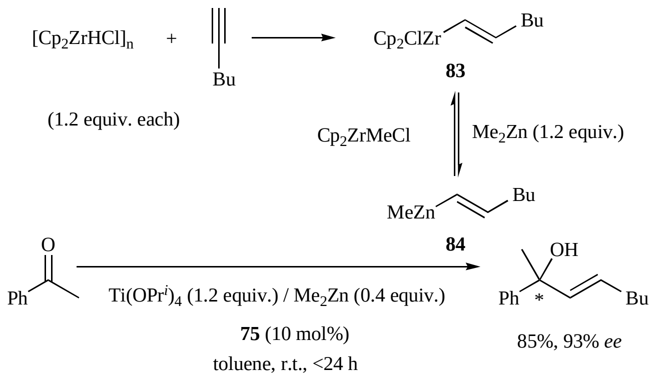



**Scheme 24.** Cinchona alkaloid catalyzed aklynylation to ketones with **82**.

### 3.9. Enantioselective Vinylation of Ketones

Recently, Li and Walsh reported a highly efficient catalytic enantioselective vinylation of aromatic, aliphatic, and $\alpha,\beta$-unsaturated ketones in the presence of 5–10 mol% of chiral titanium-based Lewis acid catalyst **75** [144]. A variety of alkynes can be employed for the *in situ* preparation of the vinylzinc species (**83** and **84**) that are suitable for the asymmetric vinylation reaction with versatile ketones. The key to success is a combination of hydro-zirconation/transmetallation to the organozinc reagent ($Me_2Zn$). The corresponding tertiary allylic alcohols are successfully obtained within 24 h in both excellent enantio-selectivities and excellent chemical yields at room temperature (Scheme **25**).

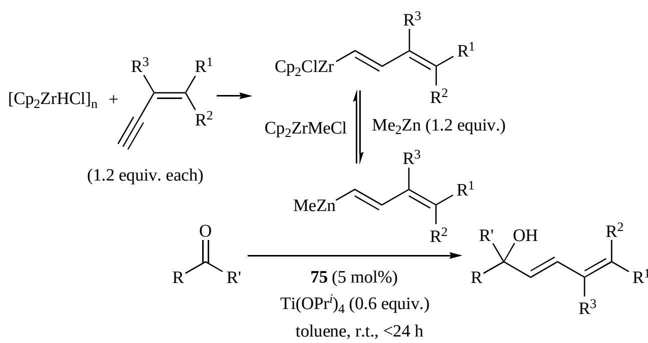
Li and Walsh also used their improved catalytic systems to explore the first catalytic enantioselective addition of dienyl groups to ketones [145]. They employed optimized reaction conditions for the vinylation reaction to the

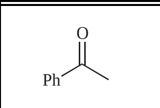
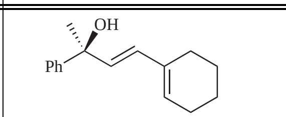
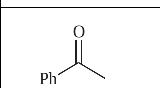
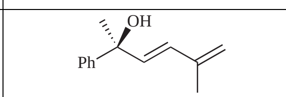
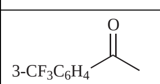
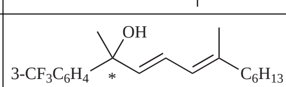
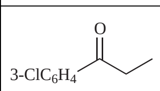
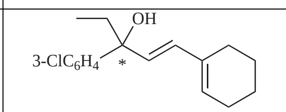
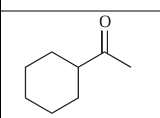
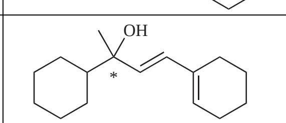



**Scheme 25.** Catalytic enantioselective vinylation to ketones with **75**.

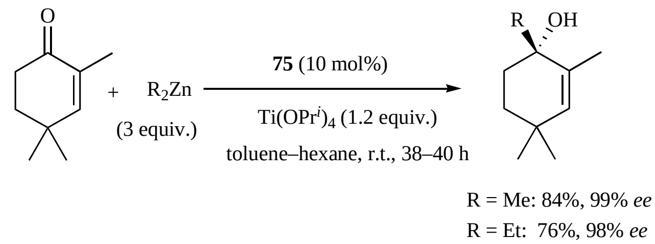
asymmetric dienyl addition and the resultant dienol products are obtained in excellent enantioselectivities and chemical yields (Table **28**). The synthetic utility of this method will increase the frontier in the category of catalytic enantioselective C–C bond forming reactions, although a considerable amount of $Ti(OPr^i)_4$ (60 mol%) is still necessary.

**Table 28. Catalytic enantioselective dienylation to ketones with 75**




| Entry | Substrate | Product | Yield (%) | *ee* (%) |
|---|---|---|---|---|
| 1 |  |  | 89 | 89 |
| 2 |  |  | 99 | 93 |
| 3 |  |  | 88 | 94 |
| 4 |  |  | 85 | 89 |
| 5 |  |  | 87 | 77 |

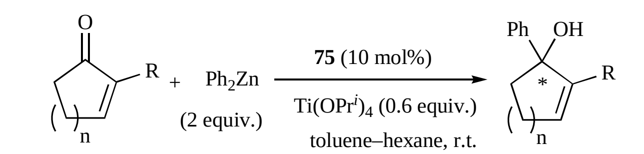
Jeon and Walsh reported that the catalytic enantioselective 1,2-addition of alkylzinc reagents with a stoichiometric amount of $Ti(OPr^i)_4$ gives cyclic α,β-unsaturated ketones (Scheme **26**) [146]. Extremely high enantioselectivities are observed for a variety of α,β-unsaturated ketones in the presence of 10 mol% of **75**, although some substrates only have moderate yields. Notably, the asymmetric catalysis with **75** is chemoselective, giving only carbonyl addition without conjugate addition. This sharply contrasts the Cu-catalyzed additions of dialkylzinc reagents to enones, which give the conjugate addition products [147].

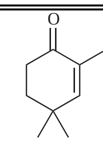
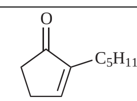
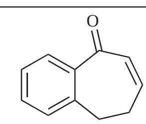
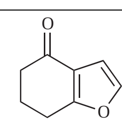
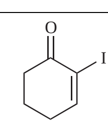



**Scheme 26.** Catalytic enantioselective 1,2-addition of alkylzinc reagents to cyclic enones with **75**.

Walsh *et al*. also established the first catalytic enantioselective addition of $Ph_2Zn$ to cyclic α,β-unsaturated ketones with 60 mol% of $Ti(OPr^i)_4$ (Table **29**) [148]. Good to excellent enantioselectivities are observed for cyclic enones with alkyl substituents in the 2-position or 2-halo substituents. The results from this study will broaden the scope of asymmetric synthesis from cyclic carbonyl compounds.

**Table 29. Catalytic enantioselective 1,2-addition of diphenylzinc to cyclic enones with 75**

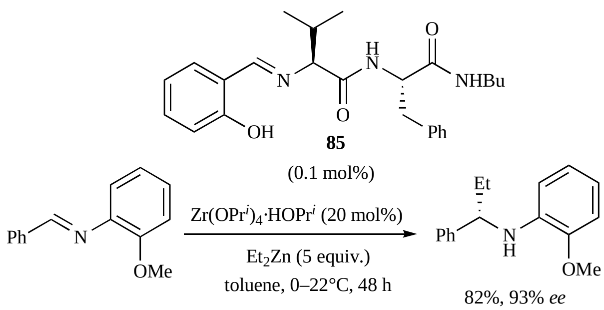



| Entry | Substrate | Yield (%) | *ee* (%) |
|---|---|---|---|
| 1 |  | 92 | 97 |
| 2 |  | 74 | 97 |
| 3 |  | 94 | 84 |
| 4 |  | 81 | 71 |
| 5 |  | 77 | 93 |

### 3.10. Enantioselective Alkylation and Arylation to Imines

Since Soai *et al*. conducted their pioneering work on the highly enantioselective addition of organozinc reagents to imines over a decade ago [149], a variety of chiral auxiliaries that are suitable for this reaction have been developed [150]. This field is rapidly progressing.

Hoveyda and Snapper *et al.* reported a highly enantioselective Zr-catalyzed addition of dialkylzinc to imines [151]. They devised chiral, peptidic Schiff base ligands that coordinate with zirconium complexes, which are optimized in a combinatorial fashion in parallel libraries. Even in the presence of 0.1 mol% of peptidic ligand **85** and 20 mol% of $Zr(OPr^i)_4{\cdot}HOPr^i$, but with excess $Et_2Zn$ (5 equiv.), asymmetric ethylation occurs smoothly with a remarkable reactivity to give the corresponding Et-adduct in 82% yield and 93% *ee* (Scheme **27**). A series of arylimines, including heterocycles, can react with a large amount of $Et_2Zn$ and practical catalyst loading (0.1–10 mol%) to give enantioenriched products in 81–97% *ee*. In this catalytic system, they found that the active chiral ligand may not be the original Schiff base **85**, but a derived amine, which is predominantly produced by the reduction of Et-Zn or Et-Zr with metal hydride *via* β-H elimination.



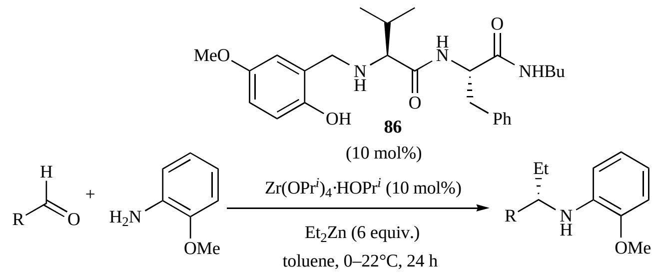
**Scheme 27.** Zr-catalyzed enantioselective Et-addition to arylimines with **85**.

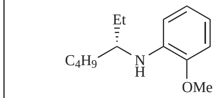
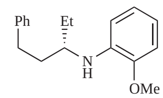
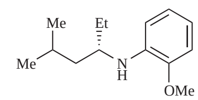
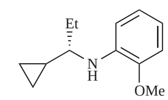
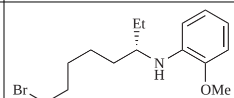
Hoveyda and Snapper *et al*. improved the chiral amine ligand and developed a three-component catalytic enantioselective synthesis for aliphatic amines where prior isolation of imines is unnecessary [152]. This is a great advantage for unstable imine starting materials, in particular aliphatic imines with acidic α-protons, which cause aldol or Mannich-type additions. The treatment of an aliphatic aldehyde and *o*-anisidine with 10 mol% of chiral, peptidic amine ligand **86** and $Zr(OPr^i)_4{\cdot}HOPr^i$ and excess $Et_2Zn$ (6 equiv.) forms the Et-added amine product in high enantiomeric excesses (Table **30**). This method is highly effective for synthesizing not only a series of chiral alkylamines (up to >98% *ee*), but also chiral arylamines (up to 91% *ee*).

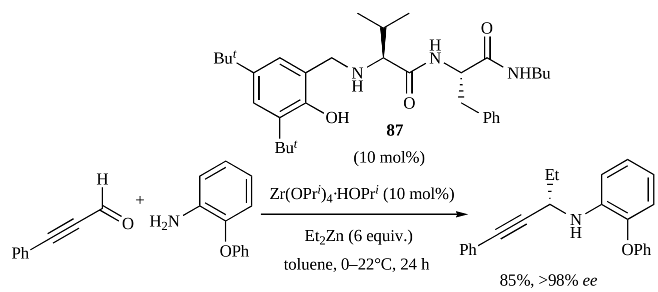
Snapper and Hoveyda *et al.* also established a three-component, enantioselective synthesis of progargylamines through a Zr-catalyzed addition of an alkylzinc reagent to alkynylimines [153]. Using 10 mol% of **87** results in a range of highly enantioenriched propargylamines in good yields, but excess diethylzinc is required (Scheme **28**).

Hoveyda and Snapper *et al.* developed a Zr-catalyzed asymmetric alkynylation [154]. 11 mol% of peptidic chiral ligand **86** in the presence of $Zr(OPr^i)_4{\cdot}HOPr^i$ can catalyze the highly enantioselective addition of dialkynylzincs, to give the corresponding optically enriched secondary propargylamines with up to 90% *ee* (Scheme **29**). Although excess bis[(trimethylsilyl)methyl]zinc is necessary to create the reactive mixed organoalkynylzinc reagents, they are critical to a smooth C–C bond formation.

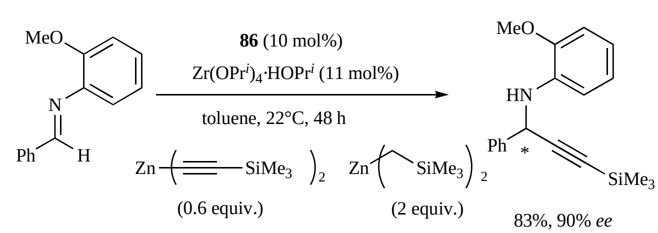
**Table 30. Three-component catalytic enantioselective synthesis of alkylamines with 86**



| Entry | Product | Yield (%) | *ee* (%) |
|---|---|---|---|
| 1 |  | 69 | 97 |
| 2 |  | >98 | 94 |
| 3 |  | 58 | 95 |
| 4 |  | 83 | 98 |
| 5 |  | 57 | 95 |



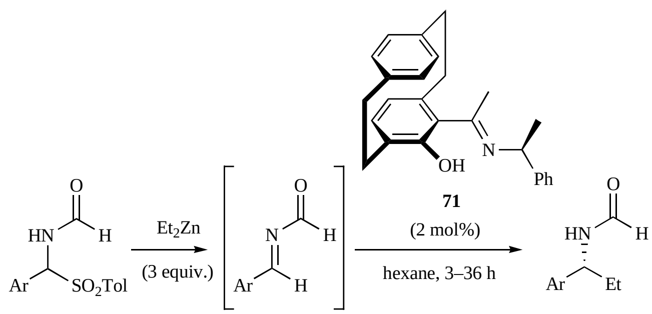
**Scheme 28.** Three-component Zr-catalyzed enantioselective alkylation to alkynylimines with **87**.



**Scheme 29.** Zr-catalyzed enantioselective alkynylation to arylimines with **86**.

Dahmen and Bräse demonstrated the first highly enantioselective diethytlzinc addition to imines in the absence of other metal catalysts such as Ti, Zr, and Cu complexes [155]. A simple method, which employs only a catalytic amount of *N*,*O*-ligand **71** based on [2,2]paracyclophane without an additional central metal, is remarkable (Table **31**). Although they applied a specific *N*-protected α-(*p*-Ts)benzylamine as the highly reactive *N*-protected imine precursors *via* an *in situ* preparation, the reaction is catalyzed by only 2 mol% of **71** for various functionalized aromatic imines. The corresponding Et-adducts are obtained in almost quantitative yields and excellent enantiomeric excesses, but 3 equiv. of diethylzinc is used.

**Table 31. *N,O*-Ligand 71 catalyzed enantioselective diethylzinc addition to imines without additional central metals**

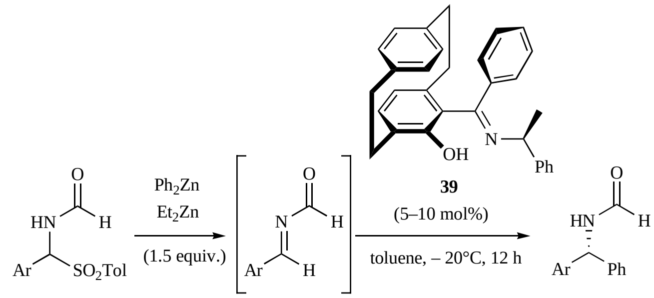



| Entry | Ar | Temp (°C) | Yield (%) | *ee* (%) |
|---|---|---|---|---|
| 1 | Ph | 10 | >99 | 95 |
| 2 | 4-$ClC_6H_4$ | 20 | >99 | 89 |
| 3 | 4-$MeOC_6H_4$ | 20 | 97 | 95 |
| 4 | 4-$(MeO_2C)C_6H_4$ | 0 | 90 | 94 |
| 5 | 2,6-$Cl_2C_6H_3$ | 0 | 98 | 95 |
| 6 | 4-$MeC_6H_4$ | 10 | >99 | 95 |

Bräse *et al.* showed that *N*,*O*-ligand **39** is highly effective for the catalytic enantioselective phenylation of imines to give diarylmethylamines [156]. For smooth phenyl transfers, a mixture of $Ph_2Zn$ and $Et_2Zn$ (1.5 equiv. each) efficiently achieves extremely high enantioselectivities (Table **32**), which are similar to Bolm's system. This catalytic system is promising, but 5–10 mol% catalytic loadings are required because of the difficulties in the catalytic enantioselective phenylation due to the high reactivities of the mixed phenylzinc reagents compared to dialkylzinc reagents.

Boezio and Charette established the first practical Cu-catalyzed system for the addition of diethylzinc reagents to *N*-diphenylphosphinoylimine [157]. The reaction proceeds extremely well with a wide range of *N*-phosphinoylimines under mild conditions in the presence of 10 mol% of $Cu(OTf)_2$ and 5 mol% of (*R*,*R*)-Me-DuPHOS (**88**) to give the Et-adducts with >90% *ee* (Table **33**). Later, Charette *et al.* improved the catalytic enantioselectivity with BozPHOS (**89**) as a bis(phosphine) monoxide ligand [158]. Remarkably, for Me and Ph additions as well as Et addition, higher enantioselectivities and reactivities are observed using **89** rather than **88,** even with a catalyst loading as low as 3 mol%. Due to these interesting results between **88** and **89**, they thoroughly examined the behavior of these ligands,

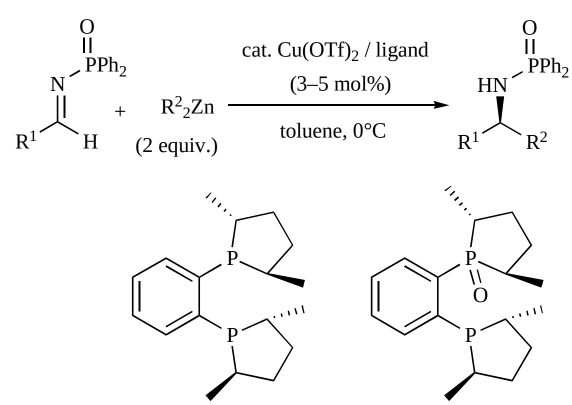
**Table 32. *N,O*-Ligand 39 catalyzed asymmetric phenyl transfer reaction to imines**




| Entry | Ar | 39 (mol%) | Yield (%) | *ee* (%) |
|---|---|---|---|---|
| 1 | 4-$MeC_6H_4$ | 5 | 99 | 94 |
| 2 | 4-$ClC_6H_4$ | 5 | 99 | 81 |
| 3 | 4-$MeOC_6H_4$ | 10 | 99 | 97 |
| 4 | 4-$Bu^tC_6H_4$ | 10 | 98 | 96 |
| 5 | 4-$(MeO_2C)C_6H_4$ | 10 | 99 | 95 |

especially oxidation under various conditions. They found that this copper-phosphine-catalyzed process initially involves the phosphine oxidation of **89** in the presence of a Cu precursor without $Et_2Zn$, which leads to a more reactive **89**-Cu-complex [159].

**Table 33. Chiral bis(phosphine)monoxide-$Cu(OTf)_2$ catalyzed enantioselective addition to imines**

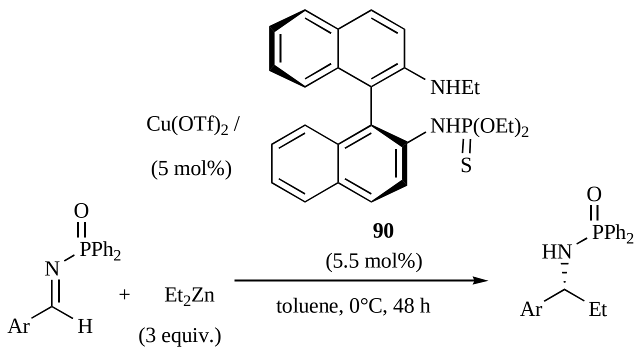



Me-DuPHOS (**88**) BozPHOS (**89**)

| Entry | $R^1$ | $R^2$ | % Yield (% *ee*) with 88 (5 mol%) | % Yield (% *ee*) with 89 (3 mol%) |
|---|---|---|---|---|
| 1 | Ph | Et | 94 (96) | 96 (98) |
| 2 | 4-$MeC_6H_4$ | Et | 91 (95) | 94 (98) |
| 3 | 4-$BrC_6H_4$ | Et | 96 (92) | 97 (97) |
| 4 | 4-$MeOC_6H_4$ | Et | 74 (95) | 91 (98) |
| 5 | 2-Furyl | Et | 97 (89) | 97 (96) |
| 6 | α-Naphthyl | Et | 93 (92) | 93 (97) |
| 7 | β-Naphthyl | Et | 94 (93) | 96 (97) |
| 8 | Cyclopropyl | Et | 82 (85) | 95 (94) |
| 9 | Ph | Me | 51 (90) | 87 (97) |
| 10 | Ph | Bu | 71 (91) | 92 (96) |

Shi and Wang reported that thiophosphoramide-$Cu(OTf)_2$ promoted catalytic enantioselective addition of diethylzinc to *N*-(diphenylphosphinoyl)imines [160]. Chiral thiophosphoramide ligand **90** with a binaphthyl backbone is used to obtain the Et-adducts in good yields and good enantioselectivities in the presence of excess diethylzinc and 5 mol% of the catalyst (Table **34**). Particularly, the presence of P=S in this ligand structure affects both the increased enantioselectivities and the decreased direct reduction by diethylzinc since P=O has a poorer performance than P=S.

**Table 34. Chiral thiophosphoramide-$Cu(OTf)_2$ catalyzed enantioselective addition to imines**




| Entry | Ar | Yield (%) | *ee* (%) |
|---|---|---|---|
| 1 | Ph | 80 | 85 |
| 2 | 2-$BrC_6H_4$ | 89 | 82 |
| 3 | 3-$MeC_6H_4$ | 89 | 82 |
| 4 | 4-$FC_6H_4$ | 83 | 85 |
| 5 | α-Naphthyl | 82 | 83 |

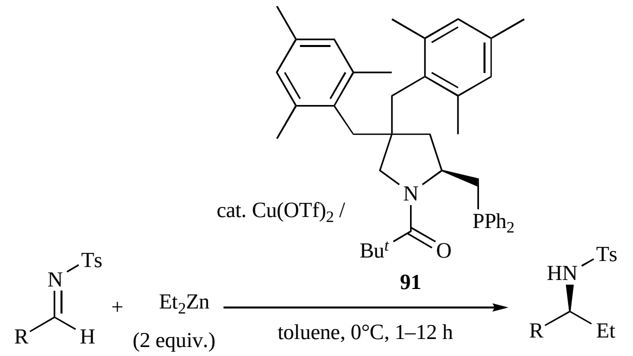
Tomioka *et al.* further improved the asymmetric alkylation using a sterically tuned chiral amidophosphine-$Cu(OTf)_2$ catalyst [161], which was previously reported in 2000 [162]. Two bulky substituents, including 2,4,6-tri-methylphenylmethyl groups on the pyrrolidine ring, are introduced in **91**. The direct reduction with diethylzinc (2 equiv.) can be minimized using 1–5 mol% of the $Cu(OTf)_2$/**91** catalyst system. The desired Et-adducts are given in high yields and high enantioselectivities (Table **35**).

Asymmetric activation, which Mikami *et al.* proposed in 1997 [77,78,163], is an effective method for the enantio-selective diethylzinc addition to imines. Gong *et al.* examined the activity of catalysts prepared by combining BINOL derivatives and chiral imines [164]. Using 10 mol% of the best combined catalyst **92**, active aromatic imines, which are prepared *in situ*, react with 3 equiv. of diethylzinc to give the corresponding adducts with high enantio-selectivities (Table **36**).

## 4. ADDITION OF ORGANOLITHIUM REAGENTS

Organolithium reagents are versatile nucleophiles in organic synthesis since they are widely available [165]. However, the catalytic enantioselective addition of organo-lithium reagents to carbonyl compounds is currently one of the most challenging C–C bond forming reactions. As organolithium reagents usually show high reactivities, it is difficult to control the amount of chiral ligands, even at low

**Table 35. Asymmetric ethylation of toluenesulfonylaldimines catalyzed by $Cu(OTf)_2$-91**

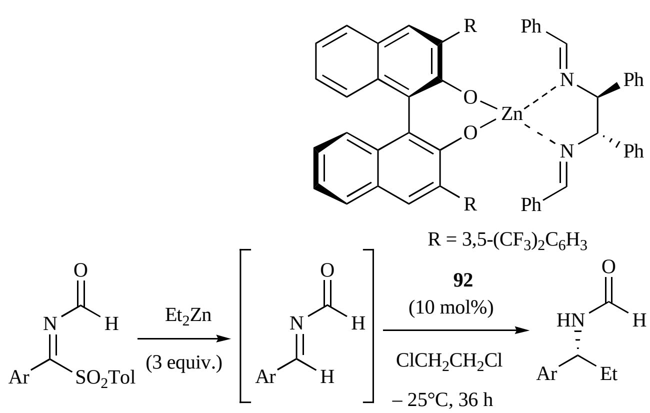



| Entry | R | $Cu(OTf)_2$/91 (mol%) | Yield (%) | *ee* (%) |
|---|---|---|---|---|
| 1 | Ph | 6.5 / 5 | 97 | 96 |
| 2 | 4-$MeOC_6H_4$ | 1.3 /1 | 78 | 86 |
| 3 | 2-Furyl | 1.3 /1 | 96 | 91 |
| 4 | Cy | 6.5 / 5 | 84 | 96 |
| 5 | $PhCH_2CH_2$ | 6.5 / 5 | 69 | 93 |

temperature. Moreover, organolithium reagents often do not tolerate a lot of functional groups, which contrasts other organometal reagents such as Zn.

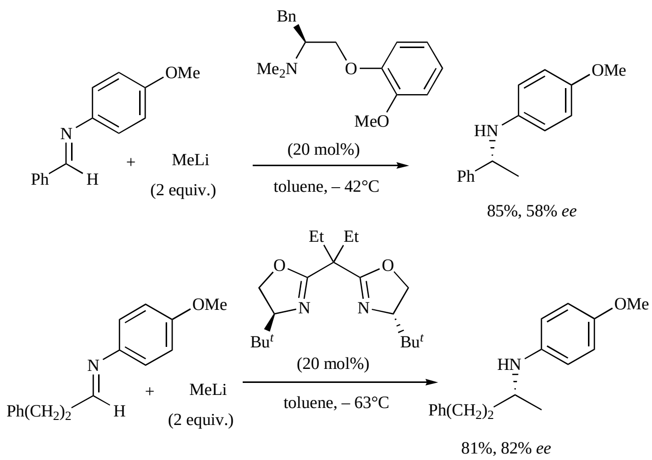
In particular, there have been difficulties in developing a direct addition of organolithium to a C=N bond in imine compounds since the basic nitrogen atoms often prevent typical chiral Lewis-acid catalysis [166]. Despite the relatively low reactivity based on the weak electrophilic nature of aldimines rather than aldehydes or ketones [167], Tomioka *et al.* in 1991 [168] and Denmark *et al.* in 1994 [169] each reported a successful enantioselective catalysis for imines with an external chiral auxiliary (Scheme **30**).

**Table 36. Asymmetric activation in catalytic enantioselective addition to imines with 92**




| Entry | Ar | Yield (%) | *ee* (%) |
|---|---|---|---|
| 1 | Ph | 91 | 83 |
| 2 | 3-$ClC_6H_4$ | 80 | 88 |
| 3 | 3-$BrC_6H_4$ | 90 | 93 |
| 4 | 4-$MeC_6H_4$ | 78 | 86 |
| 5 | 4-$NCC_6H_4$ | 75 | 94 |
| 6 | 4-$CF_3C_6H_4$ | 86 | 92 |

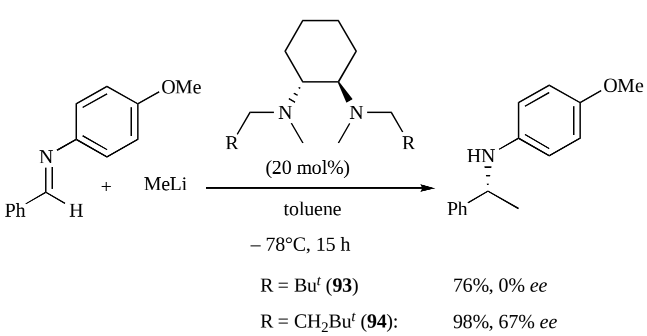
The most promising chiral ligands are bis(oxazolines) and (–)-sparteine, which can achieve enantioselectivities as high as 80% *ee* with 10–20 mol% catalyst loadings. Over the last decade, examples of enantioselective catalysis are limited, but are steadily growing. Herein, recent progress in the catalytic enantioselective addition to imines and quinilines with organolithium reagents is discussed. In these enantioselective catalysis chiral secondary or tertiary amines are obtained, which are synthetically important in the context of not only natural product chemistry, but also the fields of biology and pharmaceuticals.




**Scheme 30.** First practical, catalytic enantioselective addition to imines.

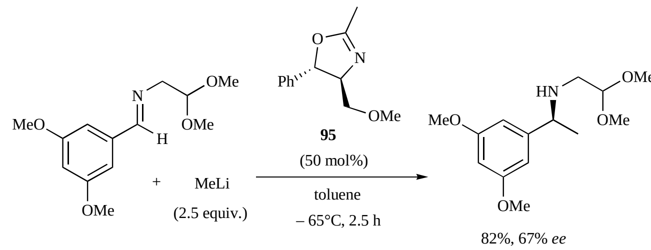
### 4.1. Enantioselective Addition to Imines

Alexakis *et al.* reported the catalytic enantioselective addition to aldimine by $C_2$-symmetric *N*,*N*-ligands based on a chiral cyclohexanediamine backbone (Scheme **31**) [170]. Several ligands are effective for the enantioselective methylation and a substituent in the β-position to the nitrogen is critical for inducing the enantioselectivity. Particularly, the reaction proceeds with 20 mol% of the chiral auxiliary (**94**) in which $(CH_2)_2Bu^t$ tethers are attached at N atoms, and the desired Me-adduct is obtained in 98% yield and 67% *ee.* Interestingly, shorter tether such as $(CH_2)_2Bu^t$ in **93** are quite ineffective (0% *ee*).




**Scheme 31.** Catalytic enantioselective addition with tertiary $C_2$ symmetric diamines **94**.

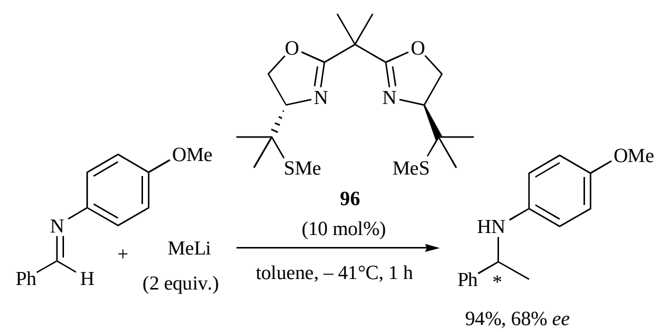
Gluszyńska and Rozwadowska adopted chiral mono-oxazoline derivatives for the catalytic enantioselective




**Scheme 32.** Enantioselective addition catalyzed by monooxazoline ligand **95**.

addition to an imine (Scheme **32**) [171]. They examined a series of chiral monooxazoline ligands on Pomeranz-Fritsch type imine with MeLi for the subsequent synthesis of isoquinoline alkaloids. The best results are obtained in the presence of excess (2.6 equiv.) chiral oxazoline **95** and give the Me-adduct in 89% yield and 76% *ee*. The catalytic reaction with 0.5 equiv. of the chiral oxazoline proceeds without a serious loss in reactivity and gives an 82% yield and 67% *ee*.

Hanessian *et al*. reported the catalytic enantioselective reaction with $C_2$-symmetrical bis(oxazolines) ligands with pendant thioether groups [172]. 10 mol% of the chiral ligand (**96**) catalyzes the enantioselective reaction with 2 equiv. of MeLi to give the product in 94% yield and 68% *ee* (Scheme **33**). However, there is not evidence that the thioether moieties are involved in a bidentate manner as a bifunctional catalyst since the asymmetric induction is equal to the *t*-Bu-bis(oxazolines) ligand.

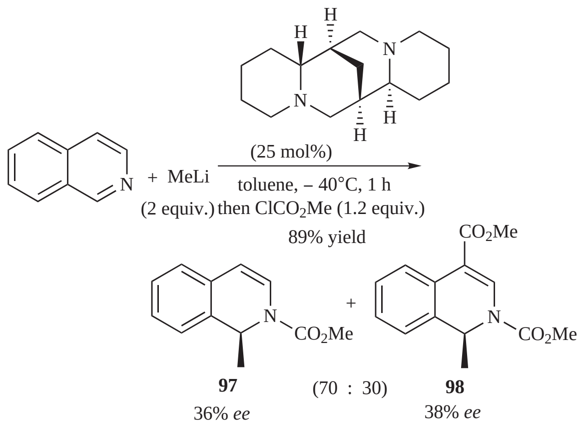



**Scheme 33.** Catalytic enantioselective addition with bis(oxazolines) ligand **96**.

### 4.2. Enantioselective Addition to Quinolines

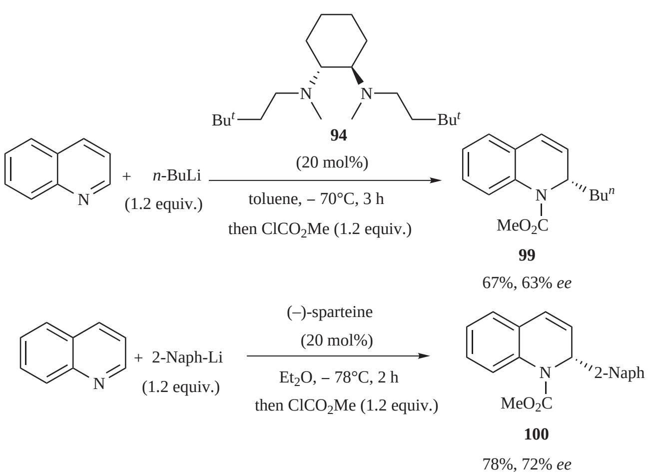
Dihydroisoquinoline derivatives are useful intermediates for synthesizing alkaloids and pharmaceutical materials. Alexakis and Amiot reported a new methodology for synthesizing chiral 1,2-dihydroisoquinolines by a direct enantioselective addition of alkyl- and aryllithium reagents to isoquinoline in the presence of (–)-sparteine [173]. Although the enantioselectivities observed are not high, the reaction proceeds with a slight excess of MeLi (1.2 equiv.) and 25 mol% of (–)-sparteine to give the mono-acylated product (**97**) in 36% *ee* along with the disubstituted product (**98**) in 38% *ee* (Scheme **34**).

Cointeaux and Alexakis also reported the catalytic enantioselective addition of organolithium reagents to quinoline [174]. $C_2$-symmetric chiral 1,2-cyclohexane-




**Scheme 34.** Catalytic enantioselective addition of methyllithium to isoquinoline with (–)-sparteine.

diamine derivatives **94** are effective in the catalytic enantioselective *n*-BuLi addition to quinoline and give the butyl adduct (**99**) in 67% yield and 63% *ee* after acylation (Scheme **35**). Moreover, (–)-sparteine (20 mol%) is highly effective for the direct enantioselective addition of 2-naphthyllithium (1.2 equiv.), and the acylated product (**100**) after a subsequent work-up. The corresponding acylated product is obtained in 78% yield and 72% *ee*.




**Scheme 35.** Catalytic enantioselective addition of organolithium reagents to quinoline.

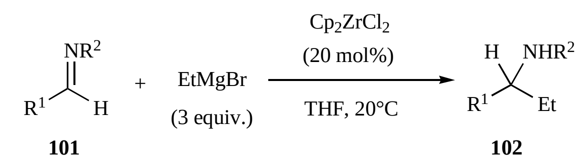
## 5. ADDITION OF ORGANOMAGNESIUM REAGENTS

The most important and traditional type of reactions using Grignard reagents (RMgX) in organic synthesis involve additions to carbonyl compounds, such as aldehydes, ketones, esters, anhydrides, imines, etc [12, 175, 176]. Compared to other popular organometal reagents such as Li and Zn, Grignard reagents are outstanding since they are commercially available as strictly titrated solutions and are easily prepared from alkyl- or aryl halides and Mg-metal. The drawback of Grignard reagents is that the reactivity is too high, which makes it difficult to control the enantioselective addition to carbonyl compounds. However, practical selective additions to carbonyl compounds with organomagnesium reagents have been recently reported.

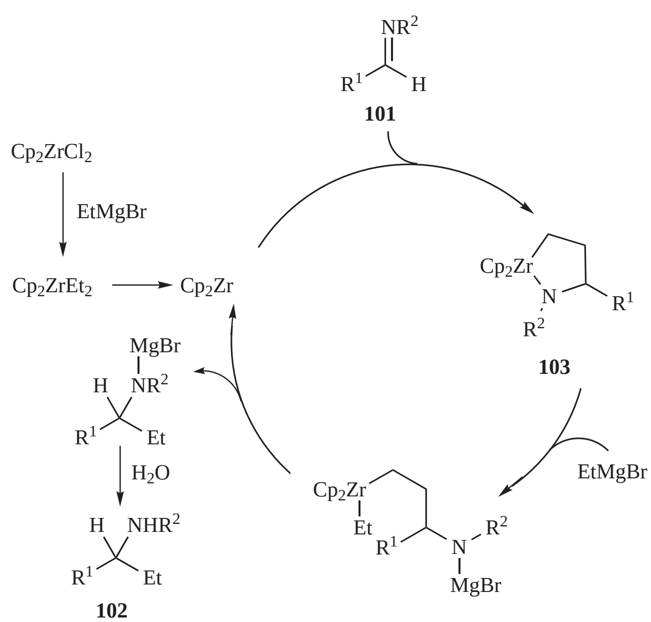
### 5.1. Addition to Imines

As we reviewed in the section of organolithium reagents, a lot of imines have little or no reactivities toward Grignard reagents due to their weak electrophilic nature. Thus, adducts are not easily formed under the usual mild reaction conditions. Although not yet applicable to asymmetric catalysis, Takahashi *et al.* found that a catalytic amount of $Cp_2ZrCl_2$ can promote the addition to aldimines (**101**) with EtMgBr to give the Et-adducts (**102**) in excellent yields [177]. Imines in Table **37** do not react with EtMgBr (3 equiv.) in THF at even 50°C. In sharp contrast, when 20 mol% of $Cp_2ZrCl_2$ is added, the reaction proceeds smoothly at room temperature and produces the corresponding amine derivatives after hydrolysis. Interestingly, the reaction of imine with a stoichiometric amount of $Cp_2Zr(CH_2{=}CH_2)$, using different Grignard reagents such as MeMgBr, EtMgBr, *n*-BuMgBr, and *sec*-BuMgBr all give the same Et-adduct **102**. Therefore, a plausible mechanism involves azazirconacyclopentane as a key intermediate (Scheme **36**): The catalytic cycle starts by the generation of an active Zr-ethylene catalyst by complexation of $Cp_2ZrCl_2$ and EtMgBr, which couples with the imine to form azazirconacyclopentane **103**. Then a subsequent ring-opening reaction with

**Table 37. Zr-catalyzed addition of EtMgBr to aldimines**




| Entry | $R^1$ | $R^2$ | Time (h) | Yield (%) |
|---|---|---|---|---|
| 1 | Ph | Bn | 1 | 96 |
| 2 | Ph | *n*-Bu | 12 | 89 |
| 3 | *t*-Bu | Bn | 1 | 97 |
| 4 | β-Naphthyl | Bn | 1 | 87 |

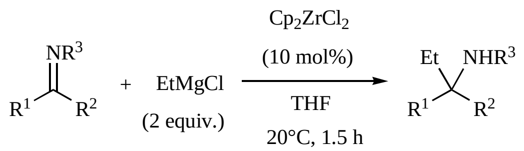



**Scheme 36.** Proposed mechanism in Zr-catalyzed EtMgBr addition to aldimines.

EtMgBr, β-elimination regenerate the active Zr-ethylene catalyst, and release the metallated amine to be hydrolyzed.

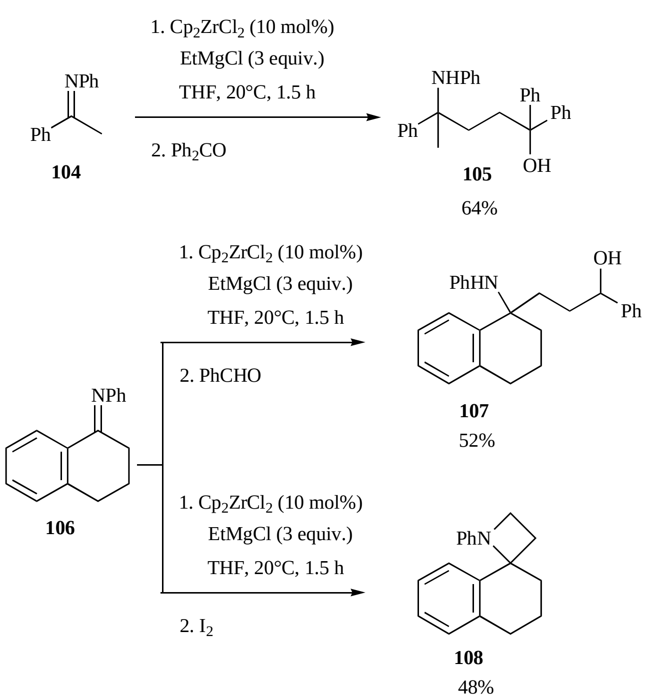
At almost the same time as independent Takahashi's work, Szymoniak *et al*. reported that a series of ketimines smoothly reacts with EtMgCl when $Cp_2ZrCl_2$ is added [178]. Both aromatic and aliphatic ketimines and benzaldimine react with EtMgCl (2 equiv.) in the presence of 10 mol% of $Cp_2ZrCl_2$ in THF at room temperature within 1.5 h to afford the corresponding amines in good yields (Table **38**).

**Table 38. Zr-catalyzed addition of EtMgCl to ketimines**




| Entry | $R^1$ | $R^2$ | $R^3$ | Yield (%) |
|---|---|---|---|---|
| 1 | Ph | Me | Ph | 78 |
| 2 | Ph | Et | Ph | 88 |
| 3 | *n*-$C_6H_{13}$ | Me | Ph | 86 |
| 4 | *n*-$C_6H_{13}$ | Me | Bn | 50 |

Furthermore, Szymoniak *et al.* applied this Zr-catalyzed addition to imines with Grignard reagents to further trap other electrophiles [179]. A one-pot reaction of the ketimines (**104** or **106**) with EtMgBr in the presence of 10 mol% of $Cp_2ZrCl_2$ and the subsequent addition of carbonyl compounds such as aldehydes or ketones, leads to δ-amino alcohols (**105** or **107**) in moderate to good yields (Scheme **37**). Moreover, the novel and rapid construction of the azetidine ring (**108**) occurs *via* a dimagnesiation-iodine-mediated cyclization. This is the first direct synthetic method that forms azaspirocyclic compounds with an azetidine ring.




**Scheme 37.** Conversion of ketimines with Zr-catalyzed Grignard reagents.

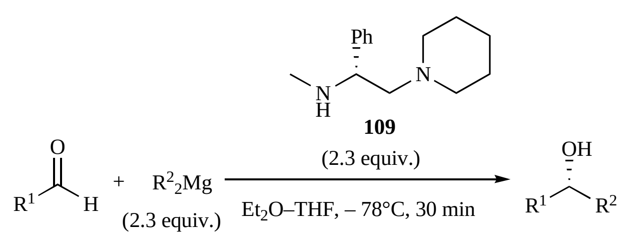
These progresses reported by Takahashi *et al.* and Szymoniak *et al.* in the Zr-catalyzed addition of Grignard reagents to aldimines and ketimines open new possibilities for synthesizing optically active amines using chiral zirconocene complexes.

### 5.2. Enantioselective Addition to Carbonyl Compounds

For C–C bond-forming reactions, the addition to carbonyl compounds with organomagnesium reagents is a popular method for producing secondary or tertiary alcohols. However, examples of the enantioselective addition with organomagnesium reagents are limited even when examples with more than an equimolar amount of chiral auxiliaries are included. Practical catalytic enantioselective additions to carbonyl compound have yet to be reported [180, 181].

Recently, Chong *et al.* reported the enantioselective alkylation of aldehydes with chiral organomagnesium amides [182]. Although excess (2.3 equiv.) chiral diamine **109** is required, chiral organomagnesium reagents prepared *in situ* from dialkylmagnesiums ($R'_2Mg$) can react with aldehydes in ether/THF at –78°C in 30 min (Table **39**). The corresponding alkyl adducts are obtained in *ca*. up to 80% *ee*. Particularly, good inductions are observed even for aliphatic aldehydes, although the yields are not high. Amine ligand **109,** which can be reused without affecting the yield or enantioselectivity for next alkylation, can be recovered at the end of the reaction by an acid-base extraction followed by distillation in >80% yield.

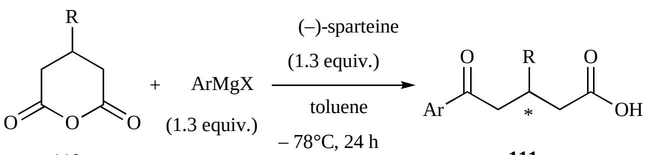
**Table 39. Enantioselective alkylation of PhCHO with chiral organomagnesium amide 109**




| Entry | $R^1$ | $R^2$ | Yield (%) | *ee* (%) |
|---|---|---|---|---|
| 1 | Ph | Et | 70 | 68 |
| 2 | Ph | *n*-Bu | 76 | 78 |
| 3 | Ph | *n*-$C_7H_{15}$ | 77 | 78 |
| 4 | Ph | *n*-$C_{10}H_{21}$ | 76 | 76 |
| 5 | 4-$MeC_6H_4$ | *n*-Bu | 57 | 76 |
| 6 | 4-$ClC_6H_4$ | *n*-Bu | 50 | 66 |
| 7 | α-Naphthyl | *n*-Bu | 71 | 54 |
| 8 | $BnO(CH_2)_7$ | *n*-Bu | 41 | 68 |

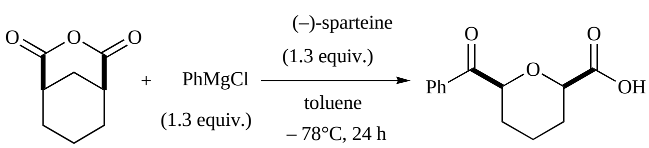
Combining alkyllithium and (–)-sparteine is a versatile method for enantioselective C–C bond forming reactions, but examples that exhibit remarkable selectivities with (–)-sparteine-Grignard reagents have yet to be reported. Shintani and Fu reported that (–)-sparteine-bound Grignard reagents desymmetrized cyclic anhydrides (**110**) to give carboxylic

**Table 40. Enantioselective desymmetrization of anhydrides with (–)-sparteine-Grignard reagents**




**110** **111**

| Entry | R | ArMgX | Yield (%) | *ee* (%) |
|---|---|---|---|---|
| 1 | Ph | PhMgCl | 91 | 92 |
| 2 | Ph | 4-MeOC$_6$H$_4$MgBr | 88 | 89 |
| 3 | Ph | 4-FC$_6$H$_4$MgBr | 82 | 78 |
| 4 | Ph | 2-MeC$_6$H$_4$MgBr | 37 | 66 |
| 5 | Bn | PhMgCl | 87 | 92 |
| 6 | *n*-Pr | PhMgCl | 74 | 90 |
| 7 | *i*-Pr | PhMgCl | 70 | 92 |
| 8 | *i*-Bu | PhMgCl | 76 | 91 |
| 9 | *t*-Bu | PhMgCl | 84 | 91 |
| 10 | 3-Thionyl | PhMgCl | 74 | 91 |




**112** **113**

77%, 82% *ee*

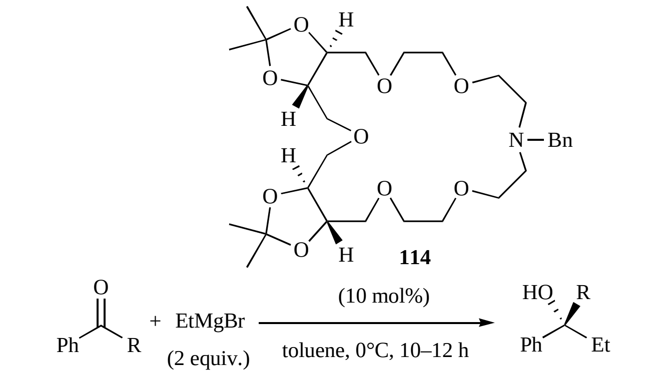
**Scheme 38.** Enantioselective desymmetrization of bicyclic anhydride with (–)-sparteine-Grignard reagents.

acids (**111**) as ring-opening products with excellent stereocontrol up to 92% *ee* [183]. For a range of anhydrides with alkyl, aryl, and heteroaromatic groups, C–C bond-forming/ring-opening reactions with a slight excess of (–)-sparteine-bound Grignard reagents proceed within 24 h in toluene at –78°C with high enantioselectivities and chemical yields (Table **40**). Not only monocyclic anhydrides, but bicyclic anhydride **112** react with a PhMgCl/(–)-sparteine reagent to generate chiral cyclic ketoacid **113** in 77% yield and 82% *ee* (Scheme **38**).

Luk'yanenko *et al.* reported the catalytic enantioselective ethylation with Grignard reagent (EtMgBr) in the presence of chiral crown ethers, which are derived from TADDOLs [184]. The most effective chiral auxiliary examined is **114** (10 mol%), and the catalytic enantioselective ethylation to benzaldehyde or acetophenone with fresh EtMgBr (2 equiv.) prepared *in situ* gives the corresponding alcohols almost quantitatively with 78% *ee* and 86% *ee*, respectively (Scheme **39**).

## 6. CONCLUSION

From the results presented in this review, the selective addition to carbonyl compounds is continually developing to meet the needs of advanced chemistry. One of the major strategies for enantioselective addition, particularly with organozinc reagents to aldehydes, is not to use other centered metals such as Ti(OPr$^i$)$_4$, which have been occasionally used in a catalytic/large amount. Other problems being addressed include high catalyst loadings and using excess organometal reagents since these problems are inconsistent with *modern* developments not only in asymmetric catalysis, but also in atom economy. Another issue is a lot of good or excellent catalysts for the well-studied alkylation to aldehydes are impractical catalysts for the alkylation of ketones or even phenylation or alkynylation of aldehydes. Except the alkylation of aldehydes, the catalytic enantioselective addition of carbonyl compounds is still a challenge and receives tremendous efforts from researchers. Thus, truly practical catalysts or methods for the selective addition to *general* carbonyl compounds using minimal amounts of traditional organometal reagents should have a significant impact on organic chemistry in the future. Moreover, we should watch future progress concerning ate complexes in organic synthesis, particularly in catalytic enantioselective synthesis.




**Scheme 39.** Asymmetric catalytic ethylation with Grignard reagent in the presence of chiral crown ethers.

## ACKNOWLEDGMENTS

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